CAMformer: A Single-Stage CNN–Transformer Hybrid for Weakly Supervised Semantic Segmentation in Aerial Imagery

Ruixue Zhou, Jihao Li, *Member, IEEE*, Wenkai Zhang, *Member, IEEE*, Shuoke Li, Jialiang Chen, *Member, IEEE*, Chongyang Li, Boyuan Tong, *Graduate Student Member, IEEE*, Weihang Zhang, *Graduate Student Member, IEEE*, Xian Sun, and Kun Fu, *Senior Member, IEEE*

***Abstract*—Weakly supervised semantic segmentation (WSSS) with image-level labels has emerged as a cost-effective solution for large-scale remote sensing (RS) imagery, yet convolutional neural network (CNN)-based approaches often suffer from limited receptive fields, incomplete activations, and background confusion in complex multiclass scenes. Transformer-based methods alleviate these issues by modeling long-range dependencies, but many existing designs are multistage and inefficient or fail to adequately address noise and contextual bias in RS data. This work proposes CAMformer, a single-stage hybrid CNN–Transformer framework that jointly captures local details and global semantics for more reliable pseudo-label generation. To further improve robustness, we introduce an interclass decoupling (ICD) strategy to resolve category conflicts, an uncertainty-weighted optimization (UWO) scheme to suppress noisy activations, and a contextual intervention sample enhancement (CISE) module to mitigate biased contextual dependencies. Extensive experiments on the challenging iSAID and ISPRS Vaihingen datasets demonstrate that CAMformer achieves state-of-the-art performance with 39.5% and 46.4% mIoU, respectively, surpassing recent one-stage methods, while cross-domain validation on PASCAL VOC2012 confirms its generalization to a certain extent. These results highlight the effectiveness of combining CNN–Transformer hybrids with context-aware learning for WSSS in RS applications.**

***Index Terms*—Class activation map (CAM), image level, remote sensing (RS), single stage, weakly supervised semantic segmentation (WSSS).**

Received 26 September 2025; revised 30 November 2025 and 23 December 2025; accepted 1 January 2026. Date of publication 6 January 2026; date of current version 30 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62331027 and Grant 62425115 and in part by the Strategic Priority Research Program of Chinese Academy of Sciences under Grant XDA0360300. *(Corresponding author: Wenkai Zhang.)*

Ruixue Zhou, Jihao Li, Shuoke Li, Jialiang Chen, Xian Sun, and Kun Fu are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Target Cognition and Application Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zhourx@aircas.ac.cn; lijh@aircas.ac.cn; lisk@aircas.ac.cn; chenjl@aircas.ac.cn; sunxian@aircas.ac.cn; fukun@aircas.ac.cn).

Wenkai Zhang, Chongyang Li, Boyuan Tong, and Weihang Zhang are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Target Cognition and Application Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zhangwk@aircas.ac.cn; lichongyang22@mails.ucas.ac.cn; tongboyuan22@mails.ucas.ac.cn; zhangweihang21@mails.ucas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2026.3651514

I. Introduction

With the rapid expansion of multisource and high-precision remote sensing (RS) data, the conventional manual and semiautomated interpretation techniques are increasingly inadequate for meeting the evolving demands of the RS industry. High spatial resolution (HSR) image interpretation has therefore become a critical tool, as it not only accelerates analysis but also improves accuracy, advancing the intelligence of RS in the big data era. Nevertheless, many current semantic segmentation approaches depend heavily on extensive, pixel-level annotated datasets, which require significant human effort and time, posing practical constraints. In response to these data constraints, recent studies in the broader RS community have actively explored advanced paradigms, such as few-shot learning with hybrid convolutional neural network (CNN)–Transformer networks [1], [2], attention-enhanced classifier designs [3], and self-supervised masked modeling for change detection [4], to maximize representational efficiency from limited samples.

To further alleviate this burden in semantic segmentation, weakly supervised learning strategies have been proposed. Among them, image-level annotations—indicating only the presence of object categories—are the easiest to obtain but also provide the least supervision, making them one of the most challenging forms in weakly supervised learning. Since Zhou et al. [5] introduced the concept of class activation maps (CAMs) for object localization, most image-level weakly supervised methods have adopted multistage frameworks [6], [7], [8]. These typically involve generating a rough localization map using a classification network, refining it into a pseudo-mask, and finally training a fully supervised segmentation model. However, such multistage pipelines require training multiple models, which complicates the workflow and reduces efficiency. To address this, several single-stage alternatives have been explored [9], [10], [11]. Yet, the majority of these approaches rely on CNNs. While CNNs excel at capturing local patterns, they are inherently constrained by limited receptive fields and struggle with modeling long-range dependencies. These limitations impair their ability to capture large-scale variations and wide spatial distributions common in RS imagery, as shown in Fig. 1(a), resulting

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

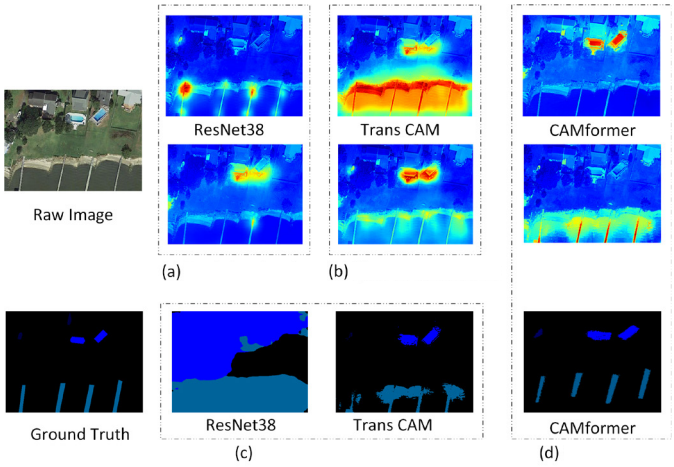


Fig. 1. Representative challenges in RS scenes. Comparisons of CAMs and segmentation results generated by different networks or methods. (a) Partial activation: conventional CNN-CAMs highlight only limited object parts, missing large regions. (b) Class co-occurrence conflict: overlapping categories confuse boundaries, causing misclassification. (c) Pseudo-mask noise: raw CAM-based pseudo-labels are semantically inconsistent and structurally discontinuous, reducing training reliability. (d) Our results: CAMformer addresses underactivation, category conflicts, and noise, yielding precise masks for robust segmentation.

in incomplete activations and reduced pseudo-mask quality. The success of Transformer models in natural language processing has inspired their rapid adoption in computer vision tasks. Thanks to their self-attention mechanism and multilayer perceptron (MLP) architecture, Transformers can model long-range semantic dependencies across spatial positions, thereby offering stronger contextual understanding than CNNs [12], [13], [14], [15], [16]. Vision Transformer (ViT) [12], as the first convolution-free Transformer for visual tasks, demonstrated that semantic-aware feature maps combined with attention significantly advance object localization. However, when applied to RS imagery, Transformers encounter new challenges: the significant differences in object sizes, the diversity of categories, and the complex background make it difficult to retain fine local details while constructing a global context, reducing the reliability of the subsequent generation of pseudo-masks. In Fig. 1(c), these challenges exacerbate the confusion between the foreground and background in complex RS scenes, and the generated pseudo-masks fail to meet the requirements of the training segmentation task.

In addition, as shown in Fig. 1(b), RS imagery often contains densely packed objects and cluttered environments where category co-occurrence is common. Under such conditions, occlusions and overlapping distributions cause prediction conflicts, while pseudo-labels derived from CAMs remain noisy and structurally fragmented. Together, these issues compromise the accuracy of weakly supervised semantic segmentation (WSSS) in RS applications. Recent Transformer-based WSSS methods, such as TransCAM [17] and MCTformer [6], have sought to mitigate these problems. TransCAM improves activation coverage but relies on a single token representation, which limits its ability to decouple multiple co-occurring categories. MCTformer incorporates multilabel supervision to refine activations but does not explicitly address noise suppression or contextual bias. Moreover, both methods adopt multistage pipelines, which are computationally costly and less efficient for large-scale RS imagery.

To overcome these limitations, we propose CAMformer, a single-stage hybrid CNN–Transformer framework designed for WSSS in RS. Built upon the Conformer backbone [18], CAMformer retains the CNN branch for local feature extraction while enhancing long-range semantic reasoning through a Transformer branch. More importantly, it introduces three complementary strategies tailored to RS challenges: an interclass decoupling (ICD) mechanism to resolve co-occurrence conflicts, an uncertainty-weighted optimization (UWO) scheme to suppress noisy pseudo-labels, and a contextual intervention sample enhancement (CISE) module to mitigate background interference. These innovations collectively facilitate the generation of more accurate and reliable pseudo-masks. Compared with the network structures of ResNet38 and TransCAM Fig. 1(d), our method addresses the issues of insufficient activation, category conflicts, and noise, thereby enhancing the robustness and effectiveness of semantic segmentation in complex RS scenarios. Extensive experiments on the challenging iSAID and ISPRS Vaihingen datasets demonstrate that CAMformer achieves state-of-the-art performance with 39.5% and 46.4% mIoU, respectively, surpassing recent one-stage methods. Furthermore, cross-domain validation on PASCAL VOC2012 yields 72.5% mIoU, confirming its generalization capabilities.

The main contributions of this work are summarized as follows.

1) *Single-Stage Hybrid Framework:* We propose CAMformer, which unifies CNN and Transformer branches to jointly capture local details and global dependencies, achieving state-of-the-art performance in image-level WSSS for RS.
2) *Conflict-Free Pseudo-Labels:* We design an ICD method to resolve class co-occurrence conflicts and generate more reliable pseudo-masks.
3) *Noise Suppression:* We introduce a UWO strategy that adaptively refines noisy regions, reducing the gap between pseudo-labels and ground truth.
4) *Context-Aware Enhancement:* We propose a lightweight CISE module to decouple targets from biased contexts, improving robustness under diverse RS environments.

II. Related Works

To better position our work, it is necessary to review existing studies from several complementary perspectives. First, we revisit general WSSS frameworks that have laid the foundation for learning with image-level supervision. Second, we examine strategies for refining and optimizing pseudo-labels, which are crucial for mitigating noise and improving mask quality. Third, we highlight recent progress in Transformer-based models for semantic segmentation, which offer stronger global reasoning but still face adaptation issues in RS imagery. Finally, we discuss context-aware augmentation approaches that explicitly address background interference and category

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

co-occurrence. Together, these four directions provide the technical background against which our CAMformer framework is developed.

A. Weakly Supervised Semantic Segmentation

WSSS seeks to alleviate the high cost of pixel-level annotations by utilizing weaker forms of supervision, such as image-level tags, bounding boxes, scribbles, or points. Among these, image-level labels are the most widely studied due to their ease of collection and wide availability. A central paradigm in image-level WSSS is the use of CAMs to generate initial object seeds. Zhou et al. [5] first introduced CAMs to highlight class-discriminative regions under image classification supervision. However, standard CAMs tend to localize only the most discriminative parts of objects, leading to underactivation of target regions and spurious activations in background areas, which severely limits their direct use as pseudo-labels.

To mitigate this issue, various seed-expansion and refinement strategies have been developed. The seed, expand, and constrain (SEC) framework [19] formulated a multiconstraint loss to encourage the growth of object seeds while regularizing spatial smoothness. Affinity-based propagation methods, such as AffinityNet [20], explicitly modeled semantic relationships between pixels to expand sparse seeds into more coherent masks. Wei et al. [21] incorporated dilated convolutions and iterative refinement to further enhance seed completeness. More recent efforts, such as IRNet [22] and AMR [23], leverage interpixel relations and region affinity regularization to progressively reduce noise in CAM-derived pseudo-labels.

Recent advancements have focused on enhancing pseudo-label quality through novel regularization strategies. For example, progressive confidence region expansion (PCRE) [24] gradually expands segmentation masks through two modules—confidence region mask expansion (CRME) and class-prototype enhancement (CPE)—to improve accuracy and address incomplete masks in earlier methods. Additionally, contrastive learning has been employed to better align CAMs with semantic regions. Token contrast (ToCo) [25] tackles oversmoothing in ViTs by introducing a patch-ToCo module, which refines patch-token relations derived from intermediate layers, improving semantic region alignment and CAM accuracy.

Despite these advances, CAM-based WSSS methods derived from CNN backbones remain inherently limited by local receptive fields, making it difficult to capture long-range dependencies and global context in complex scenes. This limitation becomes particularly pronounced in RS imagery, where objects often exhibit large-scale variations, dense distributions, and strong co-occurrence with background classes. Consequently, recent research has begun to explore hybrid CNN–Transformer architectures or attention-based mechanisms to overcome the locality of CNNs, aiming to generate more complete and reliable pseudo-labels for weakly supervised segmentation.

B. Transformers for Semantic Segmentation

ViTs [26] have recently emerged as powerful alternatives to CNNs in various vision tasks, primarily due to their ability to model long-range dependencies through self-attention mechanisms. Unlike CNNs, which are inherently constrained by local receptive fields, ViTs directly compute global relationships across image patches, thereby offering superior capabilities in capturing holistic semantic structures. Building upon this foundation, several Transformer-based architectures have been proposed for dense prediction tasks such as semantic segmentation. For example, SETR [15] demonstrated that pure Transformer encoders can achieve competitive performance in fully supervised segmentation, while Swin Transformer [27] introduced a hierarchical shifted-window attention design that balances global context modeling with computational efficiency. These advances have firmly established Transformers as state-of-the-art backbones for semantic segmentation in natural images.

In WSSS, Transformers have also shown great promise. One key limitation of CNN-derived CAMs is their tendency to focus on only the most discriminative object parts, leading to underactivated target regions and false activations in the background. By contrast, Transformer-based representations benefit from their global receptive field, which can alleviate such deficiencies. For instance, Ru et al. [28] leveraged Transformers to enhance pseudo-label generation, significantly improving object coverage and reducing spurious activations. Similarly, token-level supervision strategies have been introduced to align class and patch tokens, thereby further refining CAM quality in weakly supervised settings. Furthermore, Yoon et al. [29] proposed class tokens infusion (CTI), which injects class tokens into the Transformer architecture to guide CAM refinement.

Moreover, hybrid models combining CNNs and Transformers have also been developed to balance local feature extraction with global context modeling. TransCAM [17] integrated a Transformer branch with a CNN backbone, utilizing attention maps from the Transformer to refine CNN-generated CAMs. Similarly, Conformer [18] employed a dual-branch architecture to capture both local and global features effectively.

Recent studies have further advanced joint global–local structural feature extraction for RS analysis. For instance, Yang et al. [30] proposed ConvFormer-CD, a hybrid architecture incorporating temporal attention to capture multiscale spatiotemporal features for change detection. Similarly, SparseFormer [31] introduced a dual-CNN expert-guided Transformer to effectively aggregate local details and global context for segmentation under sparse point annotation. In parallel, Guo et al. [32] proposed a dual-view spectral and global spatial feature fusion network for discriminative representation learning, and Duan et al. [33] developed a clustering hypergraph U-Net to capture complex topological structures in hyperspectral imagery.

Although the above Transformer-based methods [30], [31], [32], [33] have advanced global–local structural modeling, both rely on dense pixel-level annotations or explicitly constructed graph structures, making them unsuitable for scenarios where only image-level labels are available. These works collectively demonstrate the importance of combining global semantic reasoning with local structural fidelity; however, the majority of them operate under full supervision and do not address the noise, class co-occurrence, and contextual

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

interference that are intrinsic to WSSS in RS. Such hybrid networks have proven effective in scenarios where objects exhibit substantial intraclass variability and dispersed spatial distributions, as is common in multiclass RS imagery. This synergy between local and global feature modeling makes hybrid CNN–Transformer frameworks particularly well-suited for addressing the challenges of WSSS in large-scale and complex scenes.

C. Pseudo-Label Generation and Refinement

The quality of pseudo-labels is widely recognized as a critical bottleneck in WSSS. Since dense pixel-level labels are unavailable, pseudo-labels generated from CAMs inevitably suffer from incomplete object coverage, background noise, and category conflicts, which can significantly undermine the training process. Early studies relied heavily on low-level graphical models for refinement. For instance, DenseCRF [34] was extensively used as a postprocessing step to sharpen object boundaries and suppress background noise. However, while effective to some extent, such post hoc refinements are limited by their inability to correct systematic errors in the initial CAMs.

To address these limitations, subsequent works proposed learning-based pseudo-label refinement strategies. IRNet [22] introduced boundary-aware affinity learning, where inter-pixel relations were exploited to propagate object seeds more accurately. Similarly, AffinityNet [20] propagated CAM seeds across semantically related regions, while AdvCAM [35] employed adversarial erasing to iteratively discover less discriminative object parts, thus expanding pseudo-label coverage. These methods mark a shift from purely postprocessing approaches toward end-to-end optimization frameworks.

Other studies have focused on noise-aware and uncertainty-aware refinement. For instance, Kendall and Gal [36] utilized deep Bayesian networks for general uncertainty estimation, though it requires large-scale pixel-level supervised data. In [37], contrastive language-image pretraining (CLIP) prior knowledge was leveraged, but its ability to generalize to domain-shifted data, such as RS images, is limited. Su et al. [38] employed random pasting of object instances into new backgrounds to mitigate background bias, yet this approach may introduce unreasonable contextual relationships (e.g., targets that would not appear in realistic backgrounds), which hinders the network's ability to effectively learn and understand contextual information. More recently, self-training paradigms [39], [40] iteratively refine pseudo-labels through teacher–student frameworks. Lin et al. [41] introduced CLIP-ES, leveraging pretrained CLIP models to align patch-text information, significantly improving pseudo-label quality and segmentation performance without additional training. However, these methods are mostly developed and evaluated on natural images and typically operate at the image level or in a category-agnostic manner, which limits their ability to explicitly handle pixel-wise class conflicts and context-induced noise in large-scale RS scenes.

In the context of RS imagery, pseudo-label refinement becomes even more critical due to the large-scale variations, class imbalance, and dense spatial distributions typical of such data. Recent works [28], [42] in RS highlight that without robust refinement strategies, pseudo-labels may amplify dataset biases or misclassify small and rare categories. Thus, pseudo-label refinement is not only a technical necessity but also a determinant of model generalization and robustness in large-scale, multiclass environments. Collectively, these strategies—from graphical models to affinity propagation, uncertainty modeling, and self-training—demonstrate that the reliability of pseudo-labels directly governs the effectiveness of WSSS pipelines. The integration of class-disentanglement and uncertainty-guided weighting, as adopted in this work, represents a promising direction to enhance pseudo-label robustness, thereby narrowing the gap between weakly and fully supervised semantic segmentation.

III. Methodology

Further information about the proposed CAMformer framework is presented in this section. Section III-A introduces some details of the single-stage hybrid segmentation model based on Transformer. Sections III-B and III-C, respectively, introduce the generation and optimization methods of pseudo-masks and the CISE method. Section III-D mainly focuses on the loss function used in CAMformer.

A. CAMformer Network

As illustrated in Fig. 2, CAMformer is built upon the Conformer architecture and consists of a CNN branch and a Transformer branch, designed to simultaneously retain CNN's advantages in capturing local features while acquiring multiscale, long-range semantic dependencies for optimizing CAMs. This hybrid design fully leverages the complementary strengths of CNNs and Transformers, enhancing feature representation and reducing semantic discrepancies in the single-stage framework.

Following Conformer's structure, the two branches are interconnected via feature coupled units (FCUs), forming an integrated combination of local convolutional blocks, self-attention modules, and MLPs. Channel dimension and feature resolution alignment are achieved using 1×1 convolutions with upsampling/downsampling strategies, further augmented by layer normalization (LN) and batch normalization (BN) to harmonize local details and global contexts. Compared to conventional CNN-based classifiers, the dual-branch hybrid architecture introduces a dual-supervision mechanism via CNN–Transformer interactions, thereby enhancing both prediction consistency and classification accuracy.

1) CNN Branch: As illustrated in Fig. 2, the CNN branch employs a multilevel feature pyramid architecture characterized by varying resolutions, with spatial resolution decreasing progressively as network depth increases. In line with a number of prior studies utilizing CNN-based architectures, we adopt a lightweight residual network, ResNet38 [20], as the backbone. Based on the specific experimental setup, the dilation rates of the final four convolutional layers have been adjusted to [1, 2, 4, 8] in order to preserve finer details

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

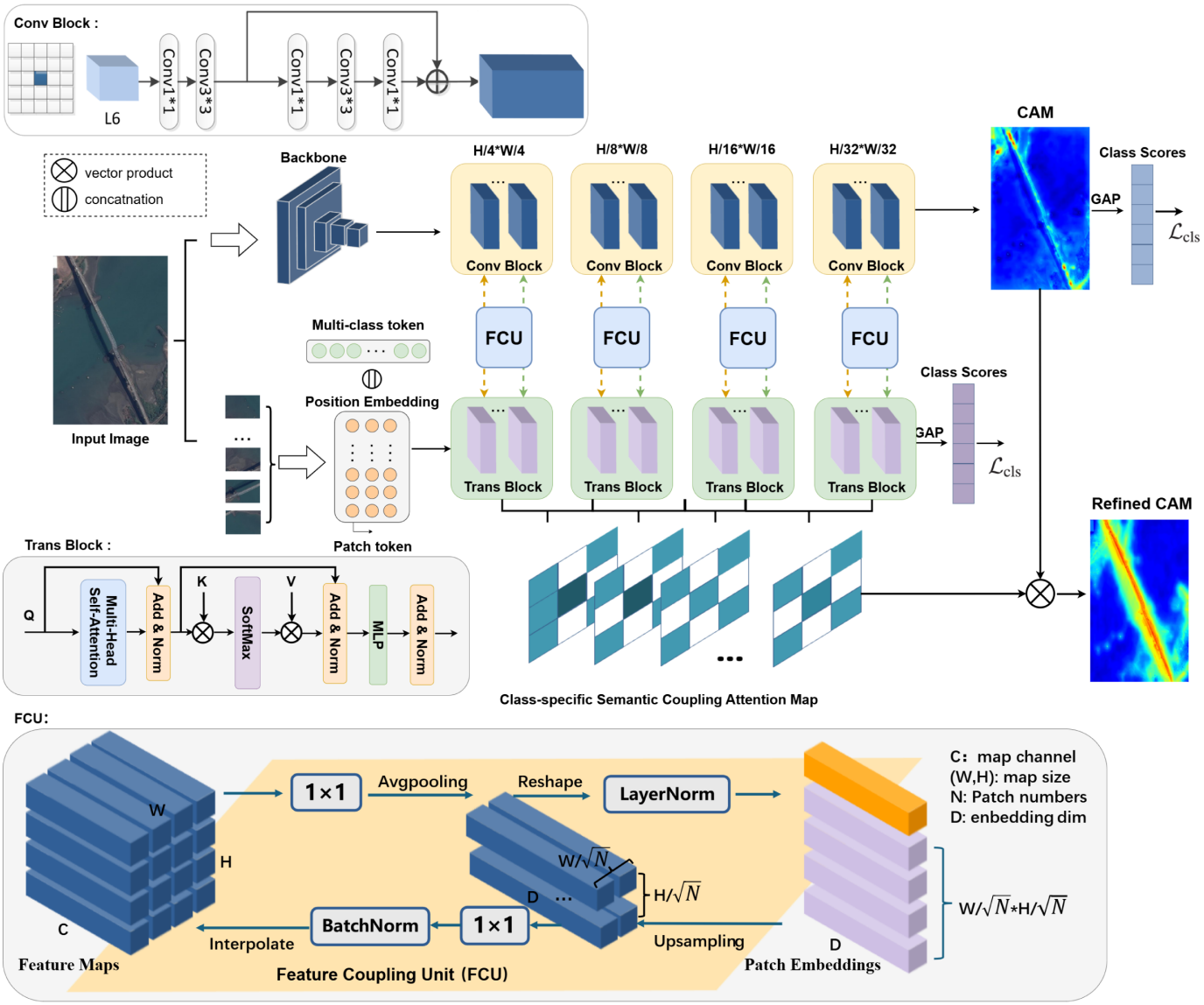


Fig. 2. Overview diagram framework of CAMformer. Overview of the proposed CAMformer framework. The network couples a CNN branch for local detail preservation with a Transformer branch for global context modeling, interconnected via FCUs. FCU is a bridge module that progressively fuses local CNN features with global transformer representations through interactive feature integration across network stages, effectively generating more complete and accurate CAMs in complex RS imagery.

of local image features. In practical applications, depending on the specific requirements of different scenarios, the backbone network can be adapted to alternative convolutional architectures, such as ResNet50, ResNet101, and ResNet152, thereby enabling flexible model expansion. To provide a more precise description of the image processing procedure within the CNN branch, this work presents the process in a formalized mathematical formulation as follows.

Given an input image $I \in \mathbb{R}^{H\times W\times 3}$ and its corresponding image label $\mathbf{Y}_{gt} \in \{0,1\}^C$, multilevel features $\mathbf{X}_l^c \in \mathbb{R}^{H\times W\times D}$ at varying resolutions are extracted through L repeated convolutional blocks ($L = 12$). Based on the feature $X_L^c(x,y)$ obtained from the final convolutional block, the foreground target feature $X_{fg}^c(x,y)$ is extracted along the channel dimension using a 1×1 convolutional layer. For the lth block

$$\mathbf{X}_{fg}^c(x,y) = \mathcal{F}_{1\times 1}\left(\mathbf{X}_L^c(x,y)\right) \quad \in \mathbb{R}^{H\times W\times C} \tag{1}$$

where $\mathbf{X}_L^c \in \mathbb{R}^{H\times W\times D}$ (input feature map); $\mathbf{X}_{fg}^c \in \mathbb{R}^{H\times W\times C}$ (output foreground features), in which $H \times W$ is the spatial resolution; D is the input channel; and $\mathcal{F}_{1\times 1}$ denotes a 1×1 convolutional layer projecting features to C class channels (C: total categories).

Subsequently, the global average pooling (GAP) layer and the fully connected (FC) layer are employed to perform classification prediction based on the foreground object features across C categories. Class scores $\mathbf{Z}^c \in \mathbb{R}^C$ are computed via

$$Z_c = \mathrm{FC}\left(\mathrm{GAP}\left(X_{fg}^c(x,y)\right)\right) \tag{2}$$

where GAP$(\cdot)$ outputs $\mathbb{R}^C$ and FC denotes an FC layer preserving dimensionality. Finally, $\mathbf{Z}^c \in \mathbb{R}^C$ is integrated with the class score obtained from the Transformer branch, and the two are weighted to produce the final class score of the entire network, which is then used in the network training process.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

2) Transformer Branch: In Fig. 2, the Transformer branch comprises multiple Transformer blocks that share the same structural design, and the total number of these blocks corresponds to the number of convolutional blocks (L). Each Transformer block primarily consists of an encoder and a decoder. The encoder includes multiple submodules that are structurally identical but do not share parameters; these are referred to as encoding layers in this work. Similarly, the decoder also contains the same number of submodules as the encoder. These submodules mainly consist of multihead self-attention mechanisms, normalization and residual connection modules, and feedforward neural networks, among others, and all of them require serialized vectors as input. Compared with the conventional Transformer architecture without convolutional components, the Transformer branch introduced in this work is capable of capturing spatial positional information among serialized input vectors without the need for explicit position embeddings. Furthermore, the integration of local feature details with global contextual representations enhances the model's feature representation capability. TransCAM, proposed by Li et al. [17], was the first work to introduce the Conformer architecture into WSSS tasks. It demonstrated that Conformer, as a backbone network, outperforms the ResNets series and achieved state-of-the-art performance on both the validation and test sets of the PASCAL VOC2012 natural scene dataset.

However, TransCAM employs only a single class token, which is insufficient to establish a one-to-one correspondence between attention mechanisms and target categories. This limitation results in the generation of category-agnostic attention maps. During the training phase, such category-agnostic attention maps inevitably cause the network to incorporate contextual information from irrelevant RS categories and complex background regions. Consequently, this introduces noise and interclass interference, which negatively impacts the precise localization of the target objects. To address the aforementioned issues, inspired by [43], we propose a multilabel-based Transformer architecture designed to generate class-aware attention (CAA) maps for the network. In contrast to previous approaches, we integrate a CNN branch as a guiding mechanism. This CNN branch not only imposes consistent classification loss constraints on both class-label predictions and patch-token classifications but also provides effective supervision regarding the local spatial positional information of class labels and patch tokens individually. Consequently, this enhancement improves the continuity and broadens the scope of target activation regions.

3) CAA Generation: Due to the inherent architectural characteristics of the Transformer, when processing image data, the input image must first be divided into unordered patch blocks and subsequently converted into patch column vectors for participation in the subsequent network training. This segmentation process results in the loss of the original spatial topological structure of the image, making it difficult to align with the features extracted by the CNN branch. Furthermore, the attention maps generated by the Transformer branch lack explicit category information, which may lead to category confusion in multiclass RS scenarios. This limitation hinders accurate semantic perception and is detrimental to the generation of precise target location maps. Therefore, this work employs the same number of class labels as the total number of categories n_{cls}. It assigns class labels to each category in order to identify the most relevant patch regions and subsequently assigns specific category labels to these matched patches to generate category-aware attention maps. The detailed procedure can be formally expressed as follows.

Given an input image $I \in \mathbb{R}^{H\times W\times 3}$, it is uniformly partitioned into N nonoverlapping patches $B \in \mathbb{R}^{h\times w\times 3}$, which are subsequently fed into the first Transformer block. Each patch B has a size of $w \times h$, where $w = W/\sqrt{N}$ and $h = H/\sqrt{N}$. These patches are then flattened and embedded into D-dimensional column vectors to form a vectorized representation of the input sequence, ultimately producing the patch tokens $B_*^1 = \{B_k^1 | k = 1, \ldots, N\}$. To facilitate the smooth execution of the classification task, it is essential to insert the initialized n_{cls} class tokens prior to the N patch tokens. Among them, the initial class labels are denoted as $C_*^1 = \{C_i^1 \mid i = 1, \ldots, n_{\text{cls}}\}$, which can be initialized randomly. The patch labels and class labels collectively constitute the input labels $t_1^* = [C_*^1 \| B_*^1]$, which are then fed into the encoder of the first Transformer block. $B_*^1 \in \mathbb{R}^{N\times D}$ denotes the patch label vector and $C_*^1 \in \mathbb{R}^{C\times D}$ denotes the class-label vector, where C, N, and D represent the spatial dimensions of the class labels and patch labels, and the embedding dimension of the Transformer, respectively. The internal processing within the encoder can be expressed as follows:

$$y_k^* = t_{k-1}^* + \text{MHA}\left(\text{LN}\left(t_{k-1}^*\right)\right) \tag{3}$$

$$t_k^* = y_k^* + \text{FFN}\left(\text{LN}\left(y_k^*\right)\right) \tag{4}$$

where MHA, LN, and FFN stand for the multihead self-attention operation, LN operation, and feedforward FC neural network, respectively. It is noteworthy that previous convolution-free Transformer network architectures typically require the addition of learnable position encodings to the vector representations of input patches in order to preserve spatial positional information during training. However, in this work, the integration of a CNN branch enables the extraction and utilization of spatial position information without the need for additional position encoding.

In each Transformer block, an encoder typically comprises multiple encoder layers that share the same structural design. Each encoder layer is composed of a multihead attention mechanism, a residual connection combined with LN, and a feedforward FC neural network. Taking the encoder of the lth Transformer block as an example, in the multihead attention mechanism, each serialized input token $X^l \in \mathbb{R}^{(C+N)\times D}$ is first projected through three distinct linear weight matrices W^Q, W^K, and W^V to produce multiple sets of query vectors $Q_i^l \in \mathbb{R}^{hw\times D_q}$, key vectors $K_i^l \in \mathbb{R}^{hw\times D_k}$, and value vectors $V_i^l \in \mathbb{R}^{hw\times D_v}$. Among them, D_q, D_k, and D_v denote the feature dimensions of the query, key, and value vectors, respectively. For a given set of vector combinations $\{Q_i^l, K_i^l, V_i^l\}$, the similarity scores between each query vector Q_i^l and all corresponding key vectors K_i^l are computed. These scores are subsequently normalized using the SoftMax function to generate a weight vector, which highlights the patch labels

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

that exhibit higher similarity to the class label. Finally, the self-attention representation S_i^l of the lth Transformer block is obtained by computing the weighted sum of the value vectors V_i^l based on the derived weight vector

$$\mathbf{S}_i^l = \text{SoftMax}\left(\frac{\mathbf{Q}_i^l \mathbf{K}_i^{l\top}}{\sqrt{D_k}}\right) \tag{5}$$

where $\mathbf{Q}_i^l, \mathbf{K}_i^l \in \mathbb{R}^{N\times D_k}$ (query/key matrices for N patches) and $\mathbf{V}_i^l \in \mathbb{R}^{N\times D_v}$ (value matrix), with $D_k = D_v = 64$ in our implementation. Among these, the SoftMax function is employed to transform the output into a probability distribution within the interval $[0, 1]$, while ^{T} denotes the transpose operation of a vector or matrix.

Based on the aforementioned, we further utilize multiple self-attention matrices, denoted as $S_i^* = [S_i^1 \| S_i^2 \| \cdots \| S_i^L]$, to extract and aggregate relevant target information. Subsequently, the output token vector of the encoder, represented as $\tilde{X}^l \in \mathbb{R}^{(C+N)\times D}$, is computed as: $\tilde{X}^l = X^l \otimes S_i^*$. Ultimately, by concatenating the output token vectors from all encoder layers, we obtain $X_t^* = [X^1 \parallel X^2 \parallel \cdots \parallel X^L]$, where L denotes the number of encoders, corresponding to the total number of Transformer blocks. Subsequently, the token vectors undergo a normalization and residual connection operation before being fed into the feedforward neural network to generate the final output of the Transformer block.

Since the feature maps are constructed by stacking multiple Transformer blocks, the self-attention mechanism S_t^l within each block captures the affinity relationships among tokens. By extracting the class-to-patch attention weights from each block, a category-aware attention map can be derived, encompassing both low- and high-level semantic representations. Considering that deeper layers tend to encode more discriminative features, we aggregate the class-to-patch attention maps $\bar{A}_t^{c2p}(L-K:L) \in \mathbb{R}^{C\times N}$ from the last K $(1 < K \le L)$ Transformer layers to obtain the final category-aware attention map

$$\bar{A}_t^{c2p}(L-K:L) = \frac{1}{K}\sum_{i=L-K}^{L} S_i^l \ [1:C, C+1:C+N] \tag{6}$$

where $\bar{A}_t^{c2p}(L-K:L)$ denotes the fused class-to-patch attention that explicitly models semantic relationships between class tokens and image patches. Here, $[1:C, C+1:C+N]$ denotes the submatrix of S_i^l from rows 1 to C and columns $C+1$ to $C+N$. The optimal choice of K is empirically determined in the subsequent experiments on the iSAID dataset.

4) Category Activation Maps Generation: In RS imagery, the inherent multiangle and multiscale acquisition often leads to significant variations in object sizes. As a result, CNN-based CAMs suffer from limited receptive fields, causing target regions to be either underactivated or excessively expanded, thereby degrading the quality of pseudo-mask generation. While existing methods [21], [44] attempt to enlarge receptive fields through multiscale training or dilated convolutions, their effectiveness remains restricted and insufficient for the wide range of object scales in RS scenes. To overcome this limitation, our model introduces a Transformer branch that complements CNN features by flexibly expanding receptive fields and capturing long-range dependencies, while retaining local detail. This integration leads to more complete and reliable target activation, facilitating higher quality pseudo-label generation.

As discussed in Section III-A1, the CNN branch extracts multiscale foreground feature representations $X_{fg}^c(x,y)$ through a series of convolutional blocks. Following the standard pipeline of CAM-based approaches, CAMs are generated by aggregating the foreground features according to their class-specific contributions. Specifically, given the classification weight vector $\mathbf{w}^c \in \mathbb{R}^d$ corresponding to class c, the CAM for class c at spatial location (x,y) is computed as

$$\hat{M}_{:,h,w}^{fg}(x,y) = \Theta\left(\frac{W_c^{\mathrm{T}} \odot X_{fg}^c(x,y)}{\|W_c{}^{\mathrm{T}}\| \odot \|X_{fg}^c(x,y)\|}\right) \tag{7}$$

where $X_{fg}^c(x,y)$ denotes the foreground feature of the pixel located at position (x,y); W_c represents the weight of the classifier corresponding to the cth class; $\odot$ denotes the element-wise multiplication; and Θ signifies the activation function, which is specifically set to ReLU in the experiments of this work to suppress spurious activation regions. Building on this foundation, we further leverage the Transformer branch to capture class-aware global spatial affinities via the attention maps $\bar{A}_t^{c2p}(L-K:L)$. These global dependencies are then employed to refine the CAMs generated by the CNN branch, denoted as $M_{:,h,w}^{fg}(x,y)$. The aggregated class-to-patch attention $\bar{\mathbf{A}}_t^{c2p} \in \mathbb{R}^{C\times N}$ refines CAMs via

$$\bar{M}_{:,h,w}^*(x,y) = \mathcal{F}_{1\times1}\left(\bar{\mathbf{A}}_t^{c2p}\right) \cdot M_{:,h,w}^{fg}(x,y) \quad \in \mathbb{R}^{\sqrt{N}\times\sqrt{N}} \tag{8}$$

where $\bar{M}_{:,h,w}^*(x,y) \in \mathbb{R}^{\sqrt{N}\times\sqrt{N}}$ denotes the refined CAM, $\mathcal{F}_{1\times1}(\cdot)$ represents a 1×1 convolutional transformation applied to the attention maps, and $\cdot$ indicates the matrix multiplication. By integrating the complementary strengths of both branches, the refined CAMs not only preserve local structural details but also achieve a broader and more consistent target activation coverage.

B. Generation and Optimization of Pseudo-Masks

As shown in Fig. 3(a), conventional pseudo-mask generation relies on applying the Argmax operation to CAMs, thereby producing hard labels (one hot). While this approach is reasonably effective in cases without occlusion, it often fails in complex multiclass RS imagery where occlusion leads to the coexistence of multiple categories within the same spatial region, resulting in prediction conflicts. Furthermore, substantial intraclass structural variability across different images introduces uncertainty, making pseudo-masks prone to noise and fragmentation. To overcome these issues, we propose a pseudo-label generation framework based on ICD, coupled with an optimization strategy termed UWO. These two components jointly mitigate interclass conflicts and reduce noise, thereby enhancing the fidelity of pseudo-labels and narrowing the gap between pseudo-labels and ground-truth annotations.

1) Pseudo-Label Generation Through ICD: To address the challenges of multiclass co-occurrence and target occlusion in RS imagery discussed earlier, we propose a pseudo-label

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

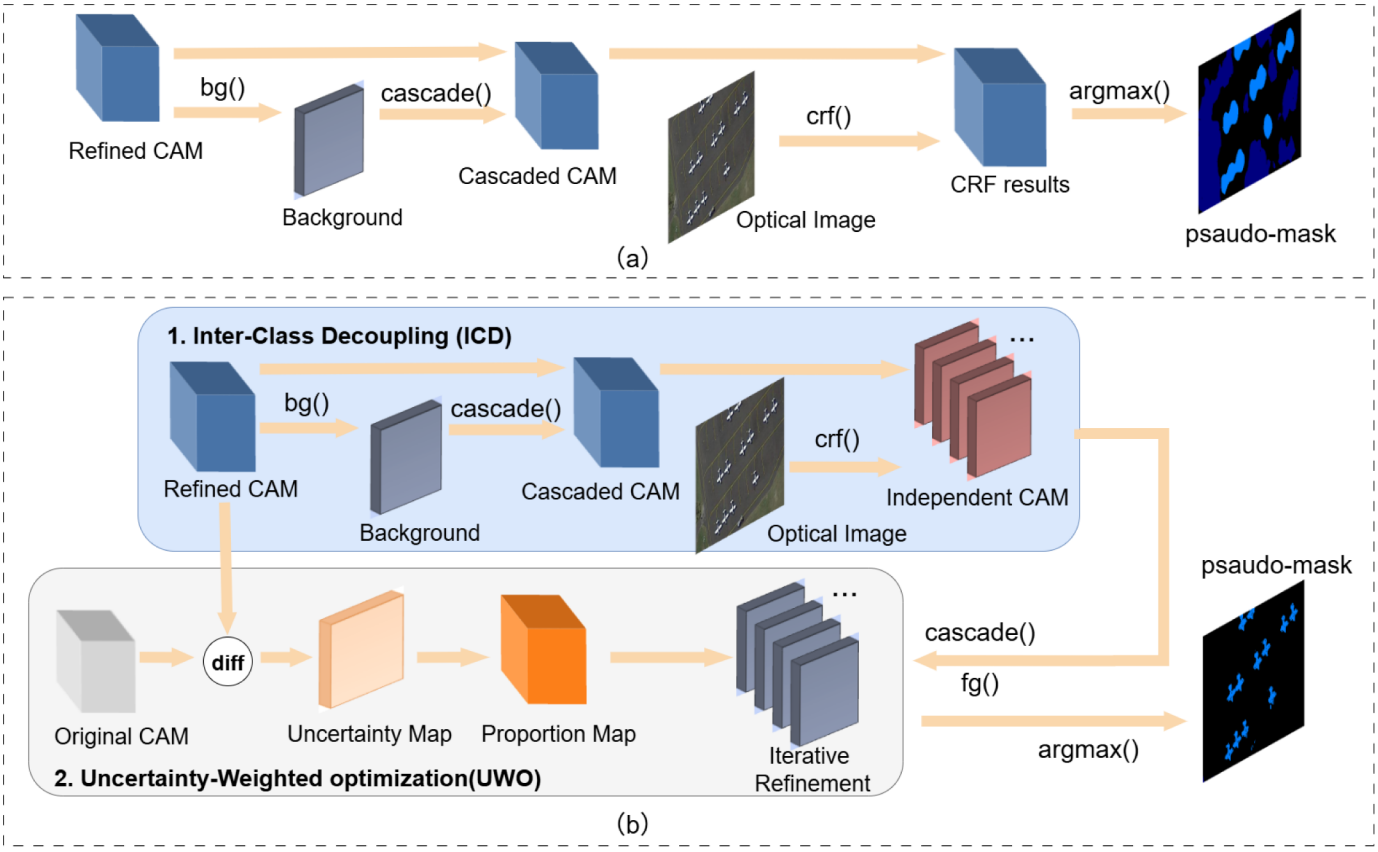


Fig. 3. Comparison of pseudo-mask generation pipeline. (a) Traditional pseudo-mask generation pipeline. The traditional CAM-based Argmax method often produces incomplete masks with category conflicts and noise. (b) Advanced pipeline of pseudo-mask generation pipeline. It consists of two stages: 1) ICD separates class features to generate conflict-free initial masks and 2) UWO leverages interbranch discrepancy to iteratively suppress noise and refine boundaries.

generation framework based on ICD. This approach first extracts category-specific features from the existing CAMs and simultaneously constructs a background-aware representation. The background-aware map is then integrated with the corresponding class features to collaboratively guide the generation of pseudo-masks. The final pseudo-mask is obtained by applying class-dependent weights to these fused features. Through this design, the ICD strategy effectively reduces pixel-level category conflicts and improves the accuracy of initial pseudo-label predictions. Formally, given the refined CAM $M^*_{:,h,w}(x,y) \in \mathbb{R}^{\sqrt{N}\times\sqrt{N}}$ derived in the preceding step, and the foreground features $X^c_{fg}(x,y) \in \mathbb{R}^{H\times W\times D}$ extracted from the backbone, the features corresponding to each category are obtained by applying $M^*_{:,h,w}(x,y)$ to the foreground feature space. For instance, the features of category i can be denoted as $\widetilde{X}^i_{fg}(x,y) \in \mathbb{R}^{H\times W\times D}$, which is computed as

$$\widetilde{X}^i_{fg}(x,y) = \mathcal{F}_{1\times 1}\left(M^*_{:,h,w}(x,y)\right) \cdot X^c_{fg}(x,y) \tag{9}$$

where $\mathcal{F}_{1\times 1}$ projects $\bar{\mathbf{M}}^*$ to match the channel dimension D of $\mathbf{X}^c_{fg}$ and $\odot$ denotes element-wise multiplication. In a similar manner, features for other categories can be extracted individually using the same procedure, ensuring that features across categories remain independent at this stage. To incorporate background information, a vector $V_{bg} \in \mathbb{R}^{H\times W\times 1}$ is randomly initialized to represent background features. The target feature $\widetilde{X}^i_{fg}(x,y)$ is concatenated with the background vector V^i_{bg}, and a 3×3 convolutional operation $\mathcal{F}_{3\times 3}(\cdot)$ is applied to compute the class score for the background feature

$$\hat{X}^{bg}_{:,h,w}(x,y) = \mathcal{F}_{3\times 3}\left(X^i_{fg}(x,y) \,\mathbb{D}\, V_{bg}\right) \tag{10}$$

where $\mathbb{D}$ denotes the channel-wise concatenation and $\mathcal{F}_{3\times 3}$ reduces the concatenated features ($\mathbb{R}^{H\times W\times(D+1)}$) to the single-channel predicted background-aware map $\hat{X}^{bg}_{:,h,w}(x,y) \in \mathbb{R}^{H\times W\times 1}$. This background-aware map is then fused with the foreground features and, together with the input image $I \in \mathbb{R}^{H\times W\times 3}$, is used to generate the pseudo-mask as follows:

$$P^i_{:,h,w}(x,y) = \text{Argmax}\left(\text{CRF}\left(I, \left[\hat{X}^{bg}_{:,h,w} \,\|\, X^i_{fg}\right]\right)\right) \tag{11}$$

where CRF$(\cdot)$ refines boundaries using image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$ and multimodal cues. Note that the CRF employed here is the standard DenseCRF. Furthermore, $P^i_{:,h,w}(x,y) \in \{0,1\}^{H\times W\times C}$ denotes a binary classification mask for category i, while the Argmax operation selects the most probable class. Final pseudo-masks $\mathbf{P} \in \{0,1\}^{H\times W\times C}$ combine per-class binary masks $\mathbf{P}^i$ with background-aware maps $\hat{\mathbf{X}}_{bg} \in \mathbb{R}^{H\times W\times 1}$. Binary masks for all categories (denoted as $\overline{P}_{:,h,w}(x,y) \in \mathbb{R}^{H\times W\times C}$) are subsequently generated and concatenated in sequence as follows:

$$\overline{P}_{:,h,w}(x,y) = P^1_{:,h,w}(x,y) \,\|\, P^2_{:,h,w}(x,y) \,\|\, \cdots \,\|\, P^C_{:,h,w}(x,y) \tag{12}$$

where $P^k_{:,h,w}(x,y)$ denote the kth component. Finally, a SoftMax function is applied to obtain a pixel-level confidence

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

map, which is then used as a weighting factor to refine the concatenated binary masks

$$\widetilde{P}_{:,h,w}(x,y) = \mathrm{SoftMax}\left(M^{*}_{:,h,w}(x,y)\right) \odot \overline{P}_{:,h,w}(x,y) \tag{13}$$

where $\odot$ denotes the element-wise multiplication. This process integrates background-aware maps into each category-specific feature derived from CAMs, thereby enriching the structural representation of the targets. It effectively alleviates underactivation and misactivation caused by substantial intraclass variations in RS data while also reducing prediction conflicts in overlapping regions. Consequently, the method enhances the reliability and overall quality of the initial pseudo-masks, providing stronger supervision for subsequent segmentation tasks.

2) Pseudo-Label Weighted Optimization Based on Uncertainty: To mitigate the noise arising from pixel-level uncertainty in RS targets within pseudo-masks, we introduce an uncertainty-guided pseudo-label optimization strategy. This approach progressively refines pseudo-labels in a self-supervised manner by emphasizing uncertain regions and incorporating local pixel color and texture cues, thereby narrowing the gap between pseudo-labels and true pixel-level annotations. The methodological details are elaborated as follows. Given the initial pseudo-mask $\widetilde{P}_{:,h,w}(x,y)$ obtained from the refined CAM $M^{*}_{:,h,w}(x,y)$, a corresponding pseudo-mask can also be generated from the original CAM $M^{fg}_{:,h,w}(x,y)$, denoted as $P_{:,h,w}(x,y)$. By comparing the pseudo-masks generated from the CNN-derived CAM and the Transformer-refined CAM, it can be observed that $\widetilde{P}_{:,h,w}(x,y)$ and $P_{:,h,w}(x,y)$ produce consistent predictions in certain regions. Under the assumption that pixels located in these overlapping areas—where both branches agree—are more likely to represent reliable, noise-free samples [45], pixel-level uncertainty is estimated by analyzing the discrepancies between the two pseudo-masks.

To enable a more reliable comparison, we adopt a thresholding strategy that introduces a parameter β to binarize the two pseudo-labels, resulting in binary confidence masks $\widetilde{M}_b(x,y)$ and $M_b(x,y)$

$$M_b(x,y) = \begin{cases} 1, & \text{if } \overline{P}_{:,h,w}(x,y) > \beta \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

$$\widetilde{M}_b(x,y) = \begin{cases} 1, & \text{if } \widetilde{P}_{:,h,w}(x,y) > \beta \\ 0, & \text{otherwise} . \end{cases} \tag{15}$$

The optimal value of β is determined experimentally in Section IV-B. Using these masks, the high-confidence regions of both $\widetilde{P}_{:,h,w}(x,y)$ and $P_{:,h,w}(x,y)$ are extracted as

$$P^{H}_{:,h,w}(x,y) = \overline{P}_{:,h,w}(x,y) \odot M_b(x,y) \tag{16}$$

$$\widetilde{P}^{H}_{:,h,w}(x,y) = \widetilde{P}_{:,h,w}(x,y) \odot \widetilde{M}_b(x,y) \tag{17}$$

where $P^{H}_{:,h,w}(x,y)$ and $\widetilde{P}^{H}_{:,h,w}(x,y)$ denote the high-confidence regions derived from the two pseudo-labels and $\odot$ denotes element-wise multiplication. The uncertainty map is then computed by measuring the discrepancy between these high-confidence predictions $\mathrm{U}(\overline{M}) \in \mathbb{R}^{H\times W\times 1}$

$$\mathrm{U}\left(\overline{M}\right) = \frac{\left\| P^{H}_{:,h,w}(x,y) - \widetilde{P}^{H}_{:,h,w}(x,y) \right\|^2}{\max\left(\left\| P^{H}_{:,h,w}(x,y) - \widetilde{P}^{H}_{:,h,w}(x,y) \right\|^2\right)} \quad \forall (x,y) \tag{18}$$

where $\|\cdot\|$ represents the $L2$ norm used to compute the mean squared error and the max function ensures normalization. To refine the initial pseudo-masks in an online manner and further reduce their deviation from ground-truth labels, uncertain pixel regions are progressively optimized. Specifically, the uncertainty map acts as a weighting factor, allowing iterative self-supervised adjustment of the pseudo-mask shape to suppress noise and improve spatial precision. During the tth iteration, the refinement process is formulated as

$$\widetilde{P}^{t}_{:,h,w}(x,y) = \sum_{(i,j)\in\mathcal{N}(h,w)} \left(\widetilde{P}^{(t-1)}_{:,h,w}(x,y) \cdot \mathrm{U}\left(\overline{M}(x,y)\right)\right) \tag{19}$$

where $\cdot$ denotes the matrix multiplication and $\widetilde{P}^{(t-1)}_{:,h,w}(x,y) \in \mathbb{R}^{H\times W\times C}$ is the pseudo-mask from the previous iteration $(t-1)$th. This iterative process reduces noise in uncertain regions and enhances the fine-grained reliability of pseudo-masks, thereby strengthening the quality of supervision for subsequent segmentation tasks.

C. Sample Augmentation Based on Contextual Intervention

Due to the wide spatial coverage of RS images, object recognition relies not only on intrinsic local features but also on surrounding objects and background conditions, which can lead to over-reliance on contextual cues. Conventional augmentation methods that expand contextual information often provide limited improvements, as they neglect critical category-specific features, especially in RS scenes with high intraclass variability. Inspired by [38], we also adopt a random pasting strategy, where object instances from source images are pasted into new backgrounds. However, unlike their method, which may introduce unreasonable contextual relationships, our approach selects pasted instances that belong to the same class as the input image, ensuring semantic coherence. Our CISE module aims to disrupt existing target–context dependencies, not to create new ones. It is lightweight and flexible, designed for easy integration into different network architectures, and experimental results confirm its effectiveness in enhancing model robustness and improving performance in complex RS environments.

To prevent occlusion between pasted instances and existing foreground objects in the host image, CAMs are employed to guide instance placement, ensuring appropriate positioning and avoiding overlap. For the iSAID dataset, single-category images are grouped into a unit-category set $I^{U}_{i} \mid i = 1,\ldots,n_U$, while multicategory images form a mixed-category set $I^{H}_{i} \mid i = 1,\ldots,n_H$, where typically $n_U \le n_H$. Given a mixed-category input I^{H}_{k} with label $Y^{H}_{k} \in \mathbb{R}^{C\times 1\times 1}$, a sample I^{U}_{k} is randomly drawn from the unit set I^{U}_{i} such that it shares at least one common class with Y^{H}_{k}. Both I^{H}_{k} and I^{U}_{k} are processed by a classification network to generate their CAMs M^{H}_{k} and M^{U}_{k}. The Argmax(·) operation is applied along the channel

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

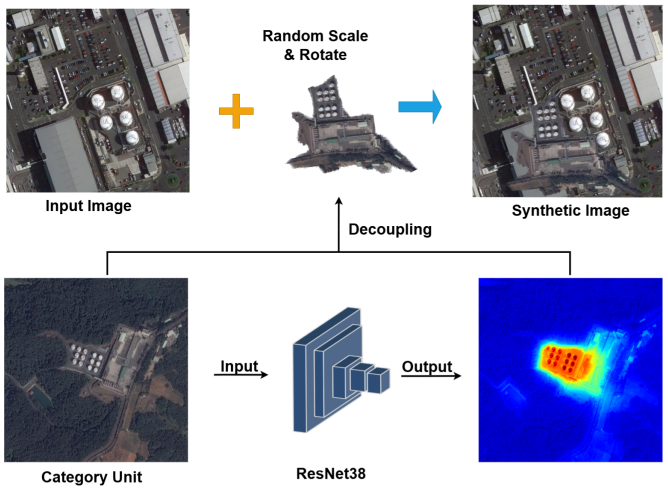


Fig. 4. Schematic of CISE module. By pasting category-consistent object instances into new backgrounds guided by CAMs, CISE decouples foreground–background dependencies and enriches contextual diversity, thereby improving pseudo-label quality and model robustness in complex RS scenes.

dimension to identify peak responses P_k^H and P_k^U, which serve as geometric centers for thresholding the CAMs via a sign function, yielding binary masks $B_i(p,q)^H$ and $B_i(p,q)^U$. The binary masks $B_i(p,q)^U$ are then used to extract the primary object regions from I_k^U. These cropped instances undergo random scaling and repositioning before being placed into the corresponding regions of I_k^H, as defined by $B_i(p,q)^H$. A reverse mean-shift operation is subsequently applied to displace pasted objects toward zero-valued background areas, avoiding overlap and preserving scene integrity. The final synthesized sample $\widetilde{I}_k^H$, therefore, contains new object instances embedded in the original contextual setting.

The synthesized images are then incorporated into the training set together with the original samples. Without the need for external datasets, this approach achieves online data augmentation and dataset expansion, exposing the model to diverse contextual variations during training. As a result, the network learns to focus more effectively on true target regions, improving both localization accuracy and discriminative capability. As shown in Fig. 4, the proposed CISE framework not only augments data diversity through CAM-guided instance pasting and geometric transformations but also introduces an online augmentation paradigm based on a twin-branch architecture. This dual-mode design breaks conventional reliance on static offline datasets, ensuring dynamic sample diversity across training epochs without increasing dataset size. By effectively disentangling foreground–background dependencies, CISE enhances model robustness in complex RS scenes, offering both methodological novelty and significant practical value.

D. Loss Function Analysis

This section details the overall loss formulation of CAMformer. The total objective $\mathcal{L}$ integrates three components: the classification loss $\mathcal{L}_{\text{cls}}$, the class-token supervision (CTS) loss $\mathcal{L}_{c\text{-token}}$, and the patch-token supervision (PTS) loss $\mathcal{L}_{p\text{-token}}$, defined as

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda\left(\mathcal{L}_{c\text{-token}} + \mathcal{L}_{p\text{-token}}\right) \tag{20}$$

where $\mathcal{L}_{\text{cls}}$ corresponds to the multilabel soft-margin classification loss, which provides the basic supervision for coarse target localization. On top of this, we introduce two complementary supervision terms: $\mathcal{L}_{c\text{-token}}$, enforcing explicit category-level guidance for class tokens, and $\mathcal{L}_{p\text{-token}}$, regularizing patch tokens to strengthen spatial consistency. Together, these losses align CNN- and Transformer-derived CAMs, establishing a dual-supervision mechanism that simultaneously expands sparse activations while preserving fine-grained object details. The coefficient λ serves as a balancing factor, adaptively regulating the contribution of the auxiliary terms to stabilize training and prevent overfitting. Further details of each loss component are provided in the following.

1) Classification (CLS) Loss: As outlined in Section III-A1, the CNN branch adopts a multiscale feature pyramid structure, where each resolution level is composed of several convolutional blocks that extract foreground target representations $X_{fg}^c(x,y)$ across different spatial scales. The extracted features are then passed through GAP followed by FC layers to produce the predicted class scores $Z_c \in \mathbb{R}^{1\times C}$. To ensure that the Transformer branch does not influence the CNN branch's classification outcomes, only the class scores from the CNN pathway are used for computing the classification loss. Consistent with standard practices in multilabel classification, the multilabel soft-margin loss is employed as the classification objective. The loss function $\mathcal{L}_{\text{cls}}$ is expressed as

$$\mathcal{L}_{\text{cls}} = -\frac{1}{C-1}\sum_{c=1}^{C-1}\left[l_c \log\sigma(Z_c) + (1-l_c)\log(1-\sigma(Z_c))\right] \tag{21}$$

where $\sigma(Z_c) = 1/(1+e^{-Z_c})$, $Z_c \in \mathbb{R}^{1\times C}$ represents the class scores predicted by the CNN branch, and $l_c \in 0,1$ denotes the corresponding ground-truth label for category c.

2) CTS Loss: As described in Section III-A2, we introduce a Transformer-based multilabel framework designed to generate class-specific attention maps. Unlike conventional Transformers, which rely on a single class label, the proposed model establishes a one-to-one correspondence between each attention head and an object category. This design enhances the Transformer branch's ability to accurately localize target regions during classification. To achieve this, the outputs of all encoder layers are concatenated, followed by a GAP operation and an FC layer to predict class scores. This process reduces the discrepancy between the class scores from the Transformer branch and those generated by the CNN branch with respect to the ground truth. Formally, the predicted class scores are computed as

$$Z_p^c = \Theta\left(\frac{\text{GAP}\left(A_t^{c2p}\times X_t^*\right)}{\max\left(\text{GAP}\left(A_t^{c2p}\times X_t^*\right)\right)}\right) \tag{22}$$

where Θ denotes the activation function (ReLU in this case), GAP refers to global average pooling, and the max operation selects the maximum value along the embedding dimension. The resulting scores are derived from token representations X_t^*, obtained by concatenating encoder outputs in the Transformer branch.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

Based on this formulation, we define the CTS loss $\mathcal{L}_{c\text{-token}}$ to align the distributions of the Transformer-predicted scores Z_p^c with the CNN branch scores Z_c, relative to the ground-truth labels l_c. Specifically, the CTS loss employs the Kullback–Leibler (KL) divergence

$$\mathcal{L}_{c-\text{token}} = \sum_{x\in\left[X^1\|X^2\|\cdots\|X^E\right]} D_{\text{KL}}^c\left(l_c \,\|\, Z_p^c(x)\right) \tag{23}$$

$$+ \sum_{x\in\left\{X_i^R, i=1,2,\ldots,L\right\}} D_{\text{KL}}^c\left(l_c \,\|\, Z_c(x)\right). \tag{24}$$

Here, D_{KL}^c measures the divergence between the predicted and true class distributions, where $D_{\text{KL}}^c = 0$ indicates perfect consistency between branches and exact alignment with ground truth. Minimizing this divergence during training enhances prediction accuracy, suppresses noisy activations, and enforces cross-branch consistency, thereby facilitating collaborative optimization of the CNN and Transformer components.

3) PTS Loss: Due to the unique structure of the Transformer, when processing image data, the input image must be divided into patches. Consequently, the spatial topological structure of the input image is disrupted. Meanwhile, the generated patch tokens lack substantial local detail information, as explicit pixel-level supervision is difficult to obtain. These issues not only hinder the predictive capability of the Transformer branch but may also adversely affect the training process of the CNN branch. Unlike the approach in [43] that generates CAMs through label-to-patch attention, we propose direct PTS. Our method first constructs interpatch attention maps and then derives a patch-based category activation map (PCAM). Specifically, we aggregate the patch-to-patch attention maps $\bar{A}_t^{p2p}(L-K:L) \in \mathbb{R}^{N\times N}$ from the last K $(1 < K \le L)$ Transformer layers to obtain the final PCAM. The overall process can be formulated as follows:

$$\bar{A}_t^{p2p}(L-K:L) = \frac{1}{K}\sum_{i=L-K}^{L} S_i^l[C+1:C+N, C+1:C+N] \tag{25}$$

$$\widetilde{M}_p^t = \text{SoftMax}\left(\mathcal{F}_{1\times1}\left(\bar{A}_t^{p2p}(L-K:L)\times X_t^*\right)\right)\odot M_{:,h,w}^{fg}(x,y) \tag{26}$$

which $\mathcal{F}_{1\times1}$ denotes a 1×1 convolution operation, which is primarily employed to adjust the dimensionality of vectors. The matrix $M_{:,h,w}^{fg}$ denotes the original CAM generated in the CNN branch. The term $S_i^l[C+1:C+N, C+1:C+N]$ represents the self-attention semantic relationships among all patch tokens within the output token X_t^*, which is derived from the Transformer branch. Here, $[C+1:C+N, C+1:C+N]$ denotes the submatrix of S_i^l from rows $C+1$ to $C+N$ and columns $C+1$ to $C+N$. These self-attention relationships are subsequently normalized using the SoftMax function, denoted as F_S, and converted into attribute values, enabling the extraction of the PCAM $\widetilde{M}_p^t$ for further processing.

Subsequently, guided by the CNN branch, we further employ the PTS loss function $L_{p\text{-token}}$ to impose explicit pixel-level constraints on the patch tokens. This approach not only provides effective supervision over the local spatial position information of patch tokens but also enhances the continuity of the appearance of the target activation regions. As a result, it further refines the granularity of the PCAM and strengthens the interaction between the Transformer branch and the CNN branch throughout the training process. The patch-marked supervised loss function $L_{p\text{-token}}$ as follows:

$$\mathcal{L}_{p\text{-token}} = \sum_{x\in\left\{\left[X^1\|X^2\|\ldots\|X^E\right]\right\}} D_{\text{KL}}^p\left(M_{:,h,w}^{fg}\|\widetilde{M}_p^t(x)\right) \tag{27}$$

which $M_{:,h,w}^{fg}$ denotes the original CAM generated in the CNN branch, whereas $\widetilde{M}_p^t$ represents the refined CAM obtained after optimization via the Transformer branch. In alignment with the class-label supervised loss formulation described previously, we employ KL divergence to quantify the distributional discrepancy between $M_{h,w}^{fg}$ and $\widetilde{M}_p^t$. This patch-level divergence metric, designated as D_{KL}^p to distinguish it from the class-wise D_{KL}^c, serves as a consistency regularization term during network optimization. Through iterative training, minimizing D_{KL}^p enforces consistency between the raw CAMs generated by the CNN branch and the PCAMs produced by the Transformer. Crucially, this process not only compels the baseline CAMs to activate additional subtle regions but also conserves their inherent capacity for local pattern detection, thereby achieving comprehensive CAM quality enhancement. To stabilize joint optimization, we implement gradient blocking during PCAM extraction, preventing CNN feature degradation while maintaining effective branch cooperation.

IV. Experiments

To comprehensively evaluate the effectiveness of the proposed framework, we adopt a two-stage experimental protocol. First, ablation studies are conducted on the challenging multiclass iSAID dataset [46] and ISPRS Vaihingen dataset [47] to quantitatively assess the contribution of individual components within CAMformer under complex variations in object scale and distribution. Second, to validate the generalization capability beyond the RS domain, we benchmark CAMformer against state-of-the-art methods on the PASCAL VOC2012 dataset [48]. The detailed experimental results and corresponding analyses are presented in this section.

A. Experimental Setup

1) Experimental Setting: Our experiments are conducted on the Conformer architecture [18], which integrates CNN branches for local feature extraction with Transformer branches for modeling multiscale long-range dependencies. Among its three variants—Conformer-Ti, Conformer-S, and Conformer-B—Conformer-S is chosen as the backbone owing to its favorable tradeoff between accuracy and computational efficiency. For the baseline, denoted as Conformer-S†, only the CNN branch is retained to produce CAMs, from which pseudo-masks are directly obtained via the Argmax operation. Training follows standard WSSS augmentation protocols [19], [20], [21], including random scaling, flipping, color jittering, and cropping. The network is optimized with the Adam optimizer (learning rate = 0.01 and weight decay = $1e^{-4}$), trained

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

for 20 epochs with a batch size of 8 on a single NVIDIA RTX 3090 GPU. During inference, we apply multiscale testing and employ DenseCRF [34] to refine pseudo labels, ensuring fair comparison with prior works.

2) Datasets and Evaluation Metrics: The iSAID dataset [46], a large-scale benchmark for aerial image segmentation, is primarily collected from JL-1 and GF-2 satellites. It consists of 15 object categories and one background class (BG), covering diverse targets such as plane (PL), baseball diamond (BD), ship (SH), ground-track field (GTF), basketball court (BC), soccer ball field (SBF), roundabout (RA), storage tank (ST), bridge (BR), large vehicle (LV), tennis court (TC), helicopter (HC), harbor (HA), small vehicle (SV), and swimming pool (SP). Abbreviations are used throughout this work for clarity. Following the official split, 1411 high-resolution images are used for training and 458 for validation. Each image is randomly cropped into 512 × 512 patches, yielding 7500 training slices, 1653 validation slices, and 1315 test slices. The ISPRS Vaihingen dataset [47] is comprised of high-resolution airborne imagery depicting a small village characterized by dense single-family houses. The dataset contains six land cover categories: impervious surfaces, low vegetation, trees, buildings, cars, and clutter. Although the original dataset provides normalized digital surface models (nDSMs), we utilize only the high-resolution orthorectified IRRG images and image-level category tags for training, excluding DSM data to maintain a strict weakly supervised setting. Following the official benchmark, 16 images are partitioned for training and the remaining 17 images for testing. The PASCAL VOC2012 dataset [48], widely recognized as a standard benchmark for detection and segmentation, contains 20 foreground categories and one background class. Following common practice, we adopt the augmented training set with additional annotations from the SBD dataset [49], yielding 10 582 training images, 1449 validation images, and 1456 testing images.

For all three datasets, only image-level category labels are used during training, representing the most challenging WSSS setting. At inference time, pixel-level annotations are employed solely for evaluation. Model performance is measured using the standard mean intersection-over-union (mIoU) metric, enabling consistent and fair comparison with prior studies. Importantly, these benchmarks complement each other: iSAID represents object-centric RS imagery with significant scale changes; ISPRS Vaihingen focuses on dense urban land cover classification; and PASCAL VOC2012 corresponds to natural scene images. By evaluating CAMformer on these diverse domains, we demonstrate not only its effectiveness in complex RS scenarios but also its generalization ability in natural image contexts.

3) Implementation Details: We adopt the Conformer-S backbone combining CNN and Transformer architectures, where the CNN branch utilizes ResNet-38 with dilation rates [1, 2, 4, 8] for multiscale feature extraction, while the Transformer branch employs a 12-layer self-attention network. Training is conducted on iSAID, ISPRS Vaihingen, and PASCAL VOC2012 datasets using the Adam optimizer (lr = 0.01 and weight decay = $1e^{-4}$) for 20 epochs with batch size 8 on an NVIDIA RTX 3090 GPU. Input images are resized to 512 × 512 and augmented via random scaling (320–640 pixels), horizontal flipping, and center cropping. The loss function combines classification loss ($\mathcal{L}_{\text{cls}}$), class-token consistency ($\mathcal{L}_{c-\text{token}}$), and patch-token consistency ($\mathcal{L}_{p-\text{token}}$) with balanced weights ($\lambda = 1$). For pseudo-label generation, we apply DenseCRF postprocessing and set the confidence threshold $\beta = 0.3$ for uncertainty weighting. During inference, we aggregate predictions from multiscale inputs (256 × 256, 512 × 512, and 768 × 768) to boost performance. All implementation strictly follows the settings described in the original CAMformer method.

TABLE I
PERFORMANCE COMPARISON OF CAMS GENERATED IN DIFFERENT BASELINE NETWORKS

Model	mIoU (%)
ResNet-38 [50]	23.6
ResNet-50 [22]	25.3
ResNet-101 [22]	26.9
Conformer − S † [18]	28.5
MCTformer − V1 [43]	29.2
MCTformer − V2 [43]	30.4
CAMformer − base	**34.7**

B. Ablation Experiments

To thoroughly evaluate the effectiveness of CAMformer, we conduct extensive ablation studies on the challenging iSAID dataset. Using Conformer-S† as the baseline, we systematically investigate the contribution of each key component, mIoU (%) and overall accuracy (OA) (%), employed as the primary evaluation metrics.

1) Baseline Setup: To ensure the rigor of the ablation experiments, we adopt a separately constructed Conformer-S† as the baseline model, where only the CNN branch is retained to generate CAMs and pseudo-masks are produced directly via the Argmax operation. As reported in Table I, Transformer-based backbones consistently outperform conventional convolutional counterparts such as the ResNet family. Specifically, the Conformer-S† baseline achieves a 1.6 percentage point improvement in mIoU over the strongest CNN backbone, ResNet-101. The MCTformer surpasses Conformer-S† by 0.7 percentage points. The full CAMformer architecture delivers the best performance, reaching an mIoU of 34.7%. It should be noted that this comparison focuses solely on backbone configurations and does not incorporate pseudo-label optimization strategies. Therefore, for clarity, we denote the backbone-only variant as CAMformer-base.

2) Parameter Settings for Transformer Branch: Given that stacking multiple Transformer blocks facilitates richer feature representations and higher layers tend to encode more discriminative and semantic information, we fuse the class-to-patch attention maps from the last K Transformer modules to construct the CAA map. To determine the optimal K, we conduct experiments with $K \in \{1, 3, 5, 7\}$ (Table II). Results show that performance improves as K increases, peaking at $K = 3$, after which excessive fusion leads to performance degradation—likely due to noise introduced by incorporating too many modules. Accordingly, $K = 3$ is adopted in the subsequent experiments.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

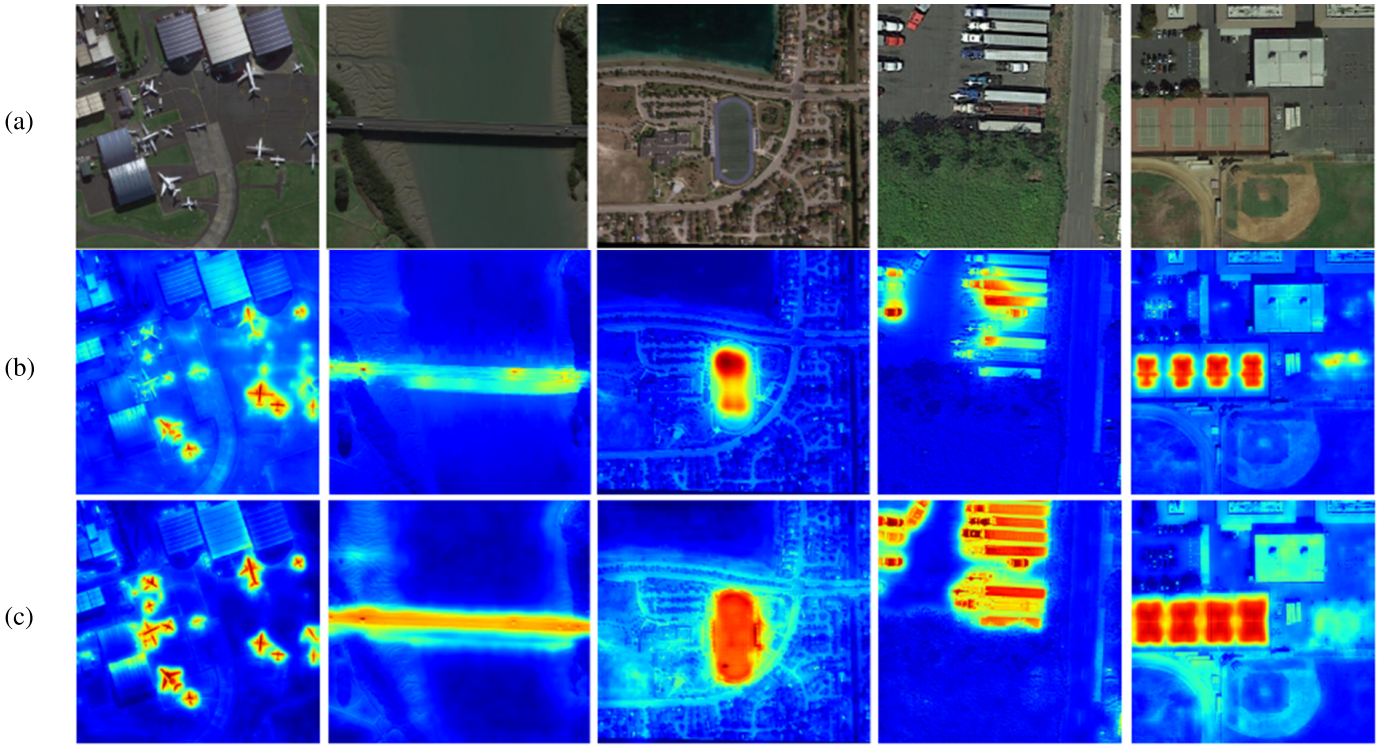


Fig. 5. Comparison of CAM visualizations before and after attention map optimization. The refined CAMs capture broader target regions and suppress background noise, yielding more complete and discriminative activations than the initial CNN-only results. (a) Original images. (b) Initial CAM. (c) Refined CAM.

TABLE II
COMPARISON RESULTS OF MIOU (%) AND OA (%) UNDER THE DIFFERENT K SETTINGS

K	mIoU (%)	OA (%)
1	28.3	80.4
3	**34.7**	**84.2**
5	32.6	83.6
7	30.5	82.7

TABLE III
ABLATION RESEARCH OF DIFFERENT COMBINATIONS OF LOSS FUNCTIONS

$\mathcal{L}_{cls}$	$\mathcal{L}_{c-token}$	$\mathcal{L}_{p-token}$	mIoU (%)
√			34.7
√	√		35.3
√		√	32.6
√	√	√	**37.1**

3) Impact of Loss Functions: In Section III-D, we investigate the effect of different combinations of three loss functions on training CAMformer: the multilabel soft-margin classification loss $\mathcal{L}_{\text{cls}}$, the CTS loss $\mathcal{L}_{c\text{-token}}$, and the PTS loss $\mathcal{L}_{p\text{-token}}$. Each of these objectives provides distinct supervision signals and contributes differently to network optimization. To evaluate their impact, we progressively add $\mathcal{L}_{c\text{-token}}$ and $\mathcal{L}_{p\text{-token}}$ on top of the baseline model trained with $\mathcal{L}_{\text{cls}}$, using mIoU (%) as the evaluation metric for CAM quality. The detailed results are summarized in Table III.

When trained solely with $\mathcal{L}_{\text{cls}}$, the baseline achieves 34.7% mIoU. Incorporating $\mathcal{L}_{c\text{-token}}$, which introduces explicit category-level supervision, improves performance to 35.3%, representing a 0.6-point gain. To isolate the contribution of the patch-token loss, we also evaluate the configuration using $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{p\text{-token}}$ only. In this case, performance degrades, indicating that patch features, which lack spatial topology and explicit semantic categories, introduce noise and semantic ambiguity when not guided by $\mathcal{L}_{c\text{-token}}$. Consequently, this loss alone has a detrimental effect on optimization. By contrast, when all three losses are jointly optimized, the network achieves the best performance, with a 2.4-point improvement in mIoU over the baseline. These results demonstrate that $\mathcal{L}_{\text{cls}}$ provides the basic classification alignment, $\mathcal{L}_{c\text{-token}}$ strengthens category-specific representation learning, and $\mathcal{L}_{p\text{-token}}$ contributes effectively only when coupled with CTS, highlighting the complementary nature of the three losses.

4) Effects of CAA: In order to further demonstrate the effect of the proposed refinement strategy, we analyze the activation behavior of CAMs before and after introducing CAA maps. Fig. 5 illustrates the representative results on five categories in RS imagery, namely, airplane, bridge, GTF, LV, and TC, which differ in scale, quantity, and spatial distribution. Prior to refinement, CAMs generated by the CNN branch alone exhibit limited activation coverage due to the restricted receptive field of convolutional operations, often leading to underactivation of target regions and spurious activations in the background. With the addition of Transformer-derived CAA maps, the self-attention mechanism effectively enlarges the receptive field and captures global semantic dependencies,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

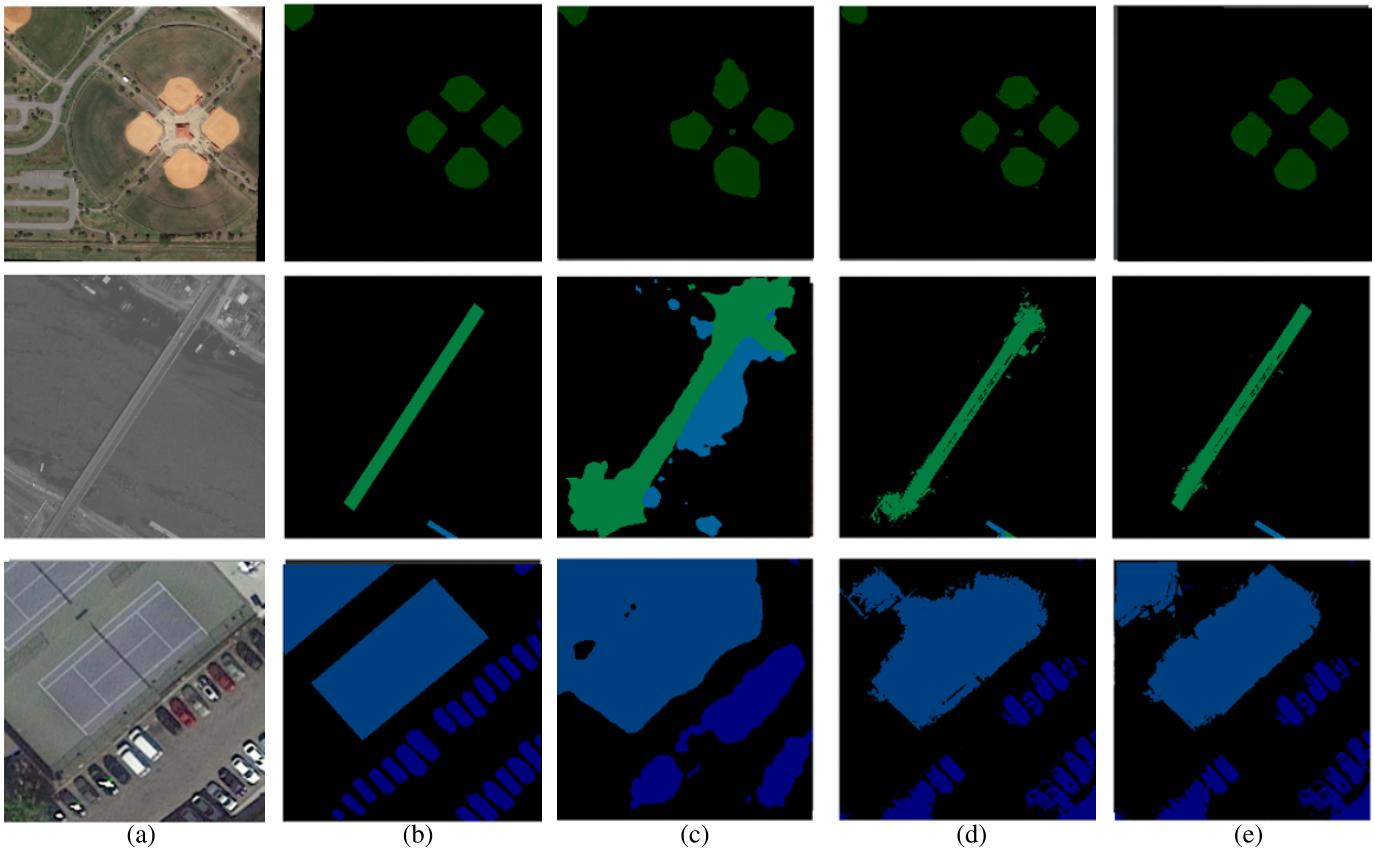


Fig. 6. Visualization effects of pseudo-mask generation and optimization strategies. (a) Input image. (b) Ground truth. (c) Initial prediction segmentation result. (d) Self-calibration pseudo-mask. (e) Final predicted segmentation result.

TABLE IV
ABLATION RESEARCH OF THE EFFECTIVENESS OF THE ICD AND UWO STRATEGY

Model	mIoU (%)	OA (%)
CAMformer-base + Argmax	32.9	81.2
CAMformer-base + ICD	36.4	84.5
CAMformer-base + Argmax + UWO	35.6	83.7
CAMformer-base + ICD + UWO	**37.1**	**85.3**

TABLE V
PERFORMANCE EVALUATION OF THE CISE STRATEGY

Model	mIoU (%)	OA (%)
Conformer – S †	30.9	81.6
Conformer – S † + CDA [40]	32.5	82.7
Conformer – S † + CISE	33.6	83.5
CAMformer w/o CISE	37.1	85.3
CAMformer w/o CISE + CDA [40]	38.2	86.0
CAMformer (Ours)	**39.5**	**86.8**

TABLE VI
COMPARISON RESULTS OF MIOU (%) AND OA (%) UNDER THE DIFFERENT β SETTINGS

β	mIoU (%)	OA (%)
0.1	34.9	82.6
0.3	**37.1**	**85.3**
0.5	36.4	84.1
0.7	35.8	83.2

enabling more complete target activation while suppressing background noise.

5) Effects of Pseudo-Mask Generation and Optimization Strategies: In Section III-B, we present two complementary strategies for improving pseudo-mask generation: an ICD method for pseudo-label construction and a UWO scheme for iterative refinement. Both techniques aim to enhance pseudo-label quality by mitigating class conflicts and correcting noisy predictions. To evaluate their effectiveness, we apply each strategy individually and in combination to the baseline CAMformer-base, assessing performance using mIoU (%) and OA (%), as reported in Table IV. For consistency, all Argmax operations include DenseCRF postprocessing, and CAMformer-base refers to the model without pseudo-label refinement and postprocessing steps. Fig. 6 shows the visual result.

The results reveal that directly applying Argmax to CAMs yields suboptimal pseudo-labels, achieving only 32.9% mIoU, which is even lower than the original CAM quality due to suppressed activations in certain categories. When the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
PERFORMANCE COMPARISON OF THE INDICATOR OF MIOU (%) ON ISAID

Model	BG	GTF	SBF	SV	SH	BR	BC	BD	RA	PL	TC	LV	ST	HA	SP	HC	mIoU
Multi-Stage weakly-supervised models.																	
AffNet [20]	77.8	14.7	43.7	3.7	5.8	6.3	18.8	20.4	9.8	7.5	54.3	23.8	42.4	3.9	1.8	1.0	20.9
IRNet [22]	78.6	18.9	39.8	6.7	16.6	7.5	24.5	16.4	5.7	15.7	54.9	29.0	29.6	9.8	2.7	3.1	22.5
CIAN [51]	78.3	21.9	40.8	8.9	17.5	8.3	28.4	22.7	13.5	18.1	47.5	27.6	30.2	8.5	4.2	5.3	23.8
SEAM [50]	73.2	27.4	36.8	4.7	14.3	6.5	41.1	30.4	15.7	19.6	43.3	30.1	23.4	7.8	9.4	8.3	24.5
SC-CAM [52]	75.2	27.6	34.1	10.1	18.0	5.7	21.1	35.9	22.2	23.8	52.3	29.1	28.8	9.3	16.3	11.9	26.6
RPNet [53]	80.2	33.6	42.1	9.8	16.9	7.2	45.2	32.6	26.1	27.6	51.5	29.4	42.7	9.3	16.8	13.7	30.3
CISM [54]	83.1	35.5	44.6	11.8	19.5	8.4	48.3	34.4	29.2	30.5	52.6	31.9	44.1	10.8	17.6	14.3	32.2
CTI [29]	83.8	37.4	46.5	11.3	18.9	9.5	50.1	36.8	32.7	33.5	55.6	32.8	47.4	13.5	20.0	15.7	34.1
TransCAM [17]	84.5	39.2	48.3	10.7	18.2	10.5	51.8	39.2	36.1	33.5	58.6	33.7	50.6	16.2	22.3	17.1	35.7
S^2EPC [55]	86.5	39.5	48.2	11.8	18.5	9.2	54.1	38.2	35.5	32.8	57.9	31.5	50.5	12.4	20.5	16.7	36.8
Single-Stage weakly-supervised models.																	
EM [56]	74.2	15.7	35.3	4.8	6.9	6.2	20.8	21.1	5.9	6.5	42.2	22.1	24.3	5.4	3.6	2.4	18.6
MIL [57]	76.9	20.1	37.6	7.5	11.3	5.8	25.2	25.7	13.8	11.3	45.1	24.5	28.9	8.1	5.9	5.5	22.1
CRF-RNN [58]	78.5	26.9	39.1	8.9	16.2	7.4	29.6	27.3	19.5	17.7	47.6	27.3	30.2	11.9	8.4	12.2	25.5
RRM [59]	80.4	30.6	40.1	10.5	20.3	8.7	32.4	29.6	26.1	26.9	51.9	30.7	34.7	12.7	12.5	15.7	28.9
1-stage [9]	83.2	32.5	44.8	16.5	26.5	9.5	32.1	33.9	34.5	35.8	49.2	34.9	31.5	14.6	19.9	17.8	31.8
AA&LR [60]	83.9	37.3	51.5	11.8	25.9	8.6	34.5	30.2	35.1	37.6	52.4	33.2	36.5	16.9	23.6	14.9	33.4
AFA [11]	84.5	33.9	54.7	12.0	29.6	9.4	37.9	32.8	32.4	33.2	57.5	37.6	44.8	14.3	22.5	20.3	34.8
SeCo [37]	88.5	34.2	56.5	13.4	31.8	10.5	41.2	34.6	35.5	35.2	59.8	35.5	44.3	17.1	26.5	20.5	36.6
FFR [61]	89.6	35.1	57.3	13.9	33.4	11.2	43.4	36.2	37.8	36.5	61.9	34.5	44.8	18.8	28.1	20.2	37.9
CAMformer	91.1	38.2	59.5	15.1	36.2	11.9	46.5	38.9	39.1	38.8	64.5	34.5	46.7	20.4	29.5	21.1	39.5

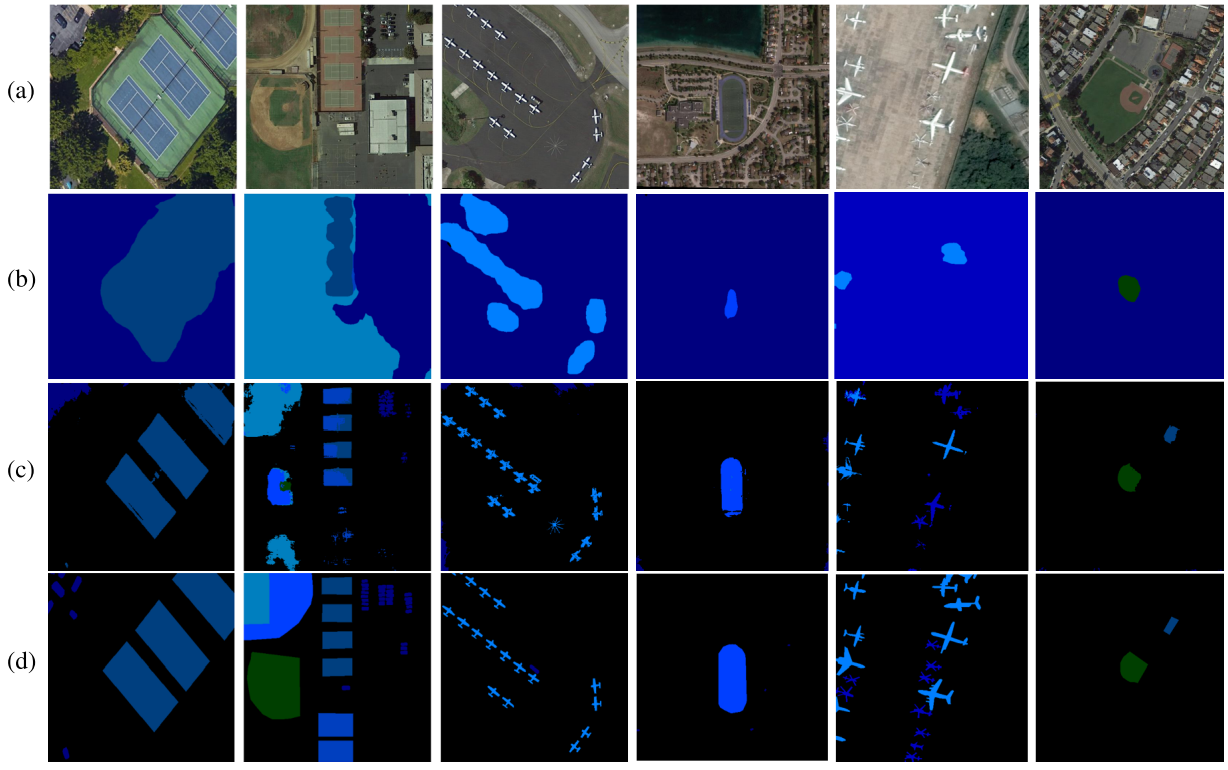


Fig. 7. Visualization of iSAID segmentation results. Compared with the baseline, CAMformer produces more complete and continuous masks with clearer boundaries and fewer background confusions, demonstrating its effectiveness in complex RS scenes. (a) Input images. (b) Baseline. (c) CAMFormer. (d) Ground truth.

ICD-based pseudo-label generation method is introduced, the suppression of activation regions in specific classes is alleviated by avoiding class conflicts at target pixels. Incorporating ICD alleviates interclass conflicts by decou-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
PERFORMANCE COMPARISON OF THE INDICATOR OF MIOU (%), OA (%), AND $F1$ (%) ON ISPRS VAIHINGEN

Model	Imp-surfs	Building	Low-Vegs	Trees	Cars	mIoU	OA	F1
Multi-Stage weakly-supervised models.								
AffNet [20]	21.5	36.3	28.4	24.9	4.6	23.1	41.1	33.0
IRNet [22]	33.4	43.8	32.6	45.2	6.8	32.3	53.4	41.7
CIAN [51]	34.5	43.2	31.5	47.2	12.6	33.8	53.6	44.5
SEAM [50]	37.5	44.2	28.5	51.5	13.3	35.0	53.9	45.9
SC-CAM [52]	46.2	51.5	40.8	50.5	18.0	41.4	54.3	53.5
RPNet [53]	45.2	51.6	39.7	52.4	18.9	41.6	59.7	51.2
CISM [54]	47.5	52.8	41.9	53.6	19.1	43.0	62.3	53.8
TransCAM [17]	49.1	53.2	43.6	54.7	19.5	44.0	63.5	54.9
CTI [29]	53.0	53.5	43.5	55.0	20.0	45.0	64.5	55.8
S^2EPC [55]	53.9	53.4	44.4	55.8	19.7	45.4	65.6	56.3
Single-Stage weakly-supervised models.								
EM [56]	22.4	32.7	26.5	23.8	5.1	22.1	42.5	32.3
MIL [57]	25.6	38.4	29.8	36.2	6.7	27.3	46.8	37.5
CRF-RNN [58]	30.2	42.1	34.5	43.7	7.9	31.7	51.2	40.8
RRM [59]	32.9	45.6	37.1	49.3	9.3	34.8	54.7	44.2
1-stage [9]	38.7	50.2	40.8	51.6	14.2	39.1	56.3	48.6
AA&LR [60]	40.5	51.9	42.4	52.8	15.7	40.7	58.1	50.3
AFA [11]	43.2	52.6	43.9	53.5	16.8	42.0	60.4	52.7
SeCo [37]	45.6	53.8	45.2	54.6	17.9	43.4	62.1	54.2
FFR [61]	47.3	54.5	46.1	55.2	18.5	44.3	63.7	55.4
CAMformer	50.5	56.2	48.1	56.8	20.4	46.4	66.5	57.4

pling category-specific activations, thereby expanding suppressed regions and significantly improving pseudo-label quality—achieving gains of 3.5 mIoU points and 3.3 OA points compared with the baseline. To isolate the effect of UWO, we apply it on top of CAMformer-base. Although the iterative optimization process increases mIoU to 35.6% and OA to 83.7%, the improvement is less pronounced than ICD, underscoring the importance of high-quality initial pseudo-label generation. Notably, combining ICD with UWO delivers the best results, with pseudo-label quality reaching 37.1% mIoU and 85.3% OA, confirming the complementarity of generation and refinement strategies.

Furthermore, we estimate uncertainty regions by comparing the discrepancies between pseudo-masks produced by the CNN and Transformer branches. To ensure fair comparison, a thresholding mechanism with parameter β is introduced to binarize confidence masks from the two pseudo-labels. Experiments with $\beta \in 0.1, 0.3, 0.5, 0.7$, summarized in Table VI, show that both mIoU and OA first increase and then decrease as β grows. The best performance is achieved at $\beta = 0.3$, highlighting the critical role of properly calibrated uncertainty thresholds in pseudo-label optimization.

6) Effects of CISE: The CISE module is designed to decouple foreground–background dependencies by introducing controlled interventions, thereby improving model robustness to diverse contextual environments. To assess its effectiveness, we compare pseudo-label quality on both the baseline model and CAMformer against the related method CDA [38]. As shown in Table V, incorporating CISE into the baseline model (Conformer – S †) yields substantial performance gains of +2.7% mIoU and +1.9% OA. Notably, CISE outperforms the competing method CDA (32.5% mIoU) by 1.1%, demonstrating its superior capability in alleviating contextual bias. Furthermore, when integrated into the CAMformer framework (which starts at 37.1% mIoU), CISE further boosts the performance to 39.5% mIoU (+2.4%) and 86.8% OA (+1.5%), significantly surpassing the configuration with CDA (38.2% mIoU). These results confirm that our proposed online intervention strategy acts as a highly effective complementary mechanism, reinforcing the discriminative capacity of the network beyond standard context decoupling techniques.

C. Comparison With Related Methods

The effectiveness of the proposed approach was comprehensively assessed on three benchmarks: the iSAID RS dataset, the challenging ISPRS Vaihingen dataset, and PASCAL VOC2012. The inclusion of the ISPRS Vaihingen dataset serves to further validate the robustness of our model in complex urban scenarios with high object density. The iSAID dataset contains 15 object categories (excluding background), with 1411 high-resolution images designated for training and validation and 458 images for testing, following the official split. To facilitate stable training of the single-stage network, images from iSAID were cropped into patches of size 512×512. The ISPRS Vaihingen dataset comprises 33 high-resolution true orthophoto (TOP) tiles covering 6 categories (impervious surfaces, buildings, low vegetation, trees, cars, and clutter). Following standard protocols, 16 tiles

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
PERFORMANCE COMPARISON OF MIOU (%) ON PASCAL VOC 2012. "SUP." DENOTES THE SUPERVISION TYPE. "I" DENOTES USING IMAGE-LEVEL SUPERVISION. "S" DENOTES USING SALIENCY MAP SUPERVISION. "R101" DENOTES RESNET101 AND "WR38" DENOTES WIDE-RESNET38. "MIT" DENOTES MIX TRANSFORMER AND "VIT" DENOTES VIT

Method	Sup.	Backbone	*val*	*test*
Multi-Stage Weakly-Supervised Models				
OAA+ [62]	*I*+*S*	R101	65.2	66.4
MCIS [63]	*I*+*S*	R101	66.2	66.9
AuxSegNet [64]	*I*+*S*	WR38	69.0	68.6
NSROM [65]	*I*+*S*	R101	70.4	70.2
EDAM [66]	*I*+*S*	WR38	70.9	70.6
EPS [67]	*I*+*S*	R101	70.9	70.8
AffNet [20]	*I*	WR38	61.7	63.7
SEAM [50]	*I*	WR38	64.5	65.7
RPNet [53]	*I*	R50	66.4	67.2
A^2GNN [68]	*I*	WR38	66.8	67.4
SC-CAM [52]	*I*	R101	66.1	65.9
CDA [38]	*I*	WR38	66.1	66.8
CISM [54]	*I*	R101	67.3	68.5
S^2EPC [55]	*I*	R101	67.6	67.8
AdvCAM [23]	*I*	R101	68.1	68.0
CPN [39]	*I*	R101	67.8	68.5
RIB [6]	*I*	R101	68.3	68.6
TransCAM [17]	*I*	Conf-S	69.3	69.6
CTI [29]	*I*	WR38	74.1	73.2
Single-Stage Weakly-Supervised Models				
EM [56]	*I*	VGG16	38.2	39.6
MIL [57]	*I*	–	42.0	40.6
CRF-RNN [58]	*I*	VGG16	52.8	53.7
RRM [59]	*I*	WR38	62.6	62.9
RRM [59]	*I*	MiT-B1	63.5	–
1Stage [9]	*I*	WR38	62.7	64.3
AA&LR [60]	*I*	WR38	63.9	64.8
AFA [11]	*I*	MiT-B1	66.0	66.3
SeCo [37]	*I*	ViT-B	74.0	73.8
FFR [61]	*I*	ViT-B	74.8	74.5
CAMformer	*I*	Conf-S	71.4	70.9
CAMformer	*I*	Conf-B	72.5	71.8

were selected for training and the remaining 17 tiles were for testing, with all tiles similarly cropped into 512×512 patches. For PASCAL VOC2012, which consists of 20 object classes and one background category, the augmented training set was constructed following standard practice, resulting in 10 582 images for training, 1449 for validation, and 1456 for testing. It is worth noting that only image-level labels were used during training for all datasets, whereas pixel-level annotations were employed solely for evaluation. Segmentation performance was measured using the mIoU metric, which is widely adopted in weakly supervised segmentation benchmarks.

1) iSAID Dataset: Table VII demonstrates that CAMformer performs comparably to advanced two-stage models on the iSAID dataset without using additional supervision during training. The results highlight its potential to advance single-stage WSSS models. However, mIoU scores for categories like bridge (BR), harbor (HA), SV, and HC are relatively lower due to dataset-specific challenges. The reduced performance can be attributed to several dataset-specific challenges: 1) bridge exhibits large intraclass scale variations, while SV represents a dense small-object category; moreover, the frequent co-occurrence of bridges and vehicles often leads to missed detections or class confusion; 2) harbor and ship frequently co-occur with the background class water, which severely interferes with foreground–background separation and object localization; and 3) HC is a sparsely represented category in the iSAID dataset, making it difficult for the network to effectively learn discriminative features. The competitive performance of CAMformer validates the effectiveness of combining Transformer and CNN in a unified framework and provides new insights for the development of efficient single-stage solutions.

Qualitative comparisons are illustrated in Fig. 7, where segmentation masks generated by CAMformer are contrasted with those of the baseline model. The baseline typically produces coarse pseudo-labels, with frequent merging of adjacent instances and large areas of background misclassification. By contrast, CAMformer achieves more precise localization of object regions and effectively suppresses background noise. Furthermore, the integration of structural priors during pseudo-label generation results in refined object boundaries that are better aligned with ground-truth masks, significantly narrowing the gap between weakly and fully supervised segmentation.

2) ISPRS Vaihingen: Table VIII shows the performance comparison of mIoU, OA, and $F1$-score on the ISPRS Vaihingen dataset. As demonstrated in the table, our proposed CAMformer achieves superior performance, setting a new state-of-the-art among single-stage weakly supervised methods with an mIoU of 46.4%, OA of 66.5%, and $F1$-score of 57.4%. Specifically, compared to recent single-stage competitors, CAMformer significantly outperforms the latest state-of-the-art method FFR [61] by 2.1% in mIoU and the Transformer-based method SeCo [37] by 3.0% in mIoU. Moreover, it surpasses widely used approaches such as AFA [11] and 1-stage [9] by substantial margins of 4.4% and 7.3% in mIoU, respectively.

It is particularly noteworthy that while maintaining the architectural simplicity and training efficiency of a single-stage framework, CAMformer achieves results that are highly competitive with, or even superior to, complex multistage

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

pipelines. For instance, our method outperforms CTI [29] (45.0% mIoU) and notably surpasses the top-performing multistage model, S^2EPC [55] (45.4% mIoU), by a margin of 1.0%. This superior performance is largely attributed to the introduction of the CISE module, which effectively mitigates contextual bias and enhances feature discrimination in complex scenes. In terms of category-specific performance, CAMformer exhibits robust segmentation capabilities on challenging classes, achieving high IoU scores on Building (56.2%) and Trees (56.8%). This validates the effectiveness of our proposed module in capturing semantic contexts and handling the scale variations inherent in high-resolution RS imagery.

3) PASCAL VOC2012 Dataset: To comprehensively assess the generalization capability of CAMformer, additional experiments were conducted on the PASCAL VOC2012 dataset. Following standard practice, 10 582 images were used for training, 1449 images were used for validation, and 1456 images were used for testing, with performance measured in terms of mIoU. As summarized in Table IX, methods are grouped by supervision type, where $\mathcal{I}$ indicates purely image-level supervision and $\mathcal{S}$ denotes the use of saliency maps as auxiliary information. CAMformer achieves results comparable to the advanced multistage method TransCAM, reaching 71.4% and 70.9% mIoU on the validation and test sets, respectively, when using Conformer-S as the backbone. It is noted that while CAMformer performs competitively, it slightly lags behind recent methods like SeCo [37] and FFR [61] on this natural image benchmark. This is primarily because our architecture—specifically the ICD and CISE modules—is specialized to resolve the dense object distribution and complex background clutter characteristic of RS imagery, whereas SeCo and FFR leverage general-purpose priors (e.g., CLIP) or frequency-domain features that are inherently more advantageous for the object-centric nature of PASCAL VOC. Nevertheless, with the more complex Conformer-B backbone, CAMformer further narrows the gap to the best-performing models. These findings confirm that the proposed model maintains robust generalization in natural scenes while excelling in its intended RS domain.

V. Conclusion

This work introduces CAMformer, a novel one-stage hybrid WSSS framework for RS imagery. By combining CNNs for local detail extraction with Transformers for modeling long-range semantic dependencies, CAMformer substantially enhances feature representation. To alleviate class conflicts in pseudo-masks caused by class co-occurrence, a class-decoupled pseudo-label generation strategy and an uncertainty-aware weighting scheme are proposed. In addition, a context intervention-based sample augmentation method is designed to improve discriminative robustness across diverse environments. Without relying on saliency cues or extra annotations, CAMformer achieves state-of-the-art weakly supervised results on large-scale RS datasets while narrowing the gap with fully supervised models and maintaining high inference efficiency.

References

[1] S. Liu, C. Fu, Y. Duan, X. Wang, and F. Luo, "Spatial–spectral enhancement and fusion network for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 5502414, doi: 10.1109/TGRS.2024.3523578.

[2] C. Fu, T. Zhou, T. Guo, Q. Zhu, F. Luo, and B. Du, "CNN-transformer and channel-spatial attention based network for hyperspectral image classification with few samples," *Neural Netw.*, vol. 186, Jun. 2025, Art. no. 107283.

[3] C. Fu, B. Du, and L. Zhang, "Do we need learnable classifiers? A hyperspectral image classification algorithm based on attention-enhanced ResBlock-in-ResBlock and ETF classifier," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505013, doi: 10.1109/TGRS.2024.3353383.

[4] T. Zhou et al., "STMNet: Single-temporal mask-based network for self-supervised hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5502712, doi: 10.1109/TGRS.2024.3523541.

[5] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

[6] J. Lee, J. Choi, J. Mok, and S. Yoon, "Reducing information bottleneck for weakly supervised semantic segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 27408–27421.

[7] J. Lee, E. Kim, J. Mok, and S. Yoon, "Anti-adversarially manipulated attributions for weakly supervised semantic segmentation and object localization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1618–1634, Mar. 2024.

[8] L. Ru, B. Du, and C. Wu, "Learning visual words for weakly-supervised semantic segmentation," in *Proc. 30th Int. Joint Conf. Artif. Intell.*, Aug. 2021, pp. 982–988.

[9] N. Araslanov and S. Roth, "Single-stage semantic segmentation from image labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4252–4261.

[10] M. Ergül and A. Alatan, "Depth is all you need: Single-stage weakly supervised semantic segmentation from image-level supervision," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2022, pp. 4233–4237.

[11] L. Ru, Y. Zhan, B. Yu, and B. Du, "Learning affinity from attention: End-to-end weakly-supervised semantic segmentation with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16846–16855.

[12] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[13] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 10347–10357.

[14] B. Wu et al., "Visual transformers: Token-based image representation and processing for computer vision," 2020, *arXiv:2006.03677*.

[15] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.

[16] L. Yuan et al., "Tokens-to-Token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 538–547.

[17] R. Li, Z. Mai, C. Trabelsi, Z. Zhang, J. Jang, and S. Sanner, "TransCAM: Transformer attention-based CAM refinement for weakly supervised semantic segmentation," 2022, *arXiv:2203.07239*.

[18] Z. Peng et al., "Conformer: Local features coupling global representations for visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 357–366.

[19] A. Kolesnikov and C. H. Lampert, "Seed, expand and constrain: Three principles for weakly-supervised image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 695–711.

[20] J. Ahn and S. Kwak, "Learning pixel-level semantic affinity with image-level supervision for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4981–4990.

[21] Y. Wei, H. Xiao, H. Shi, Z. Jie, J. Feng, and T. S. Huang, "Revisiting dilated convolution: A simple approach for weakly- and semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7268–7277.

[22] J. Ahn, S. Cho, and S. Kwak, "Weakly supervised learning of instance segmentation with inter-pixel relations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2204–2213.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

[23] J. Lee, E. Kim, and S. Yoon, "Anti-adversarially manipulated attributions for weakly and semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4071–4080.

[24] X. Xu et al., "Weakly supervised semantic segmentation via progressive confidence region expansion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 9829–9838.

[25] L. Ru, H. Zheng, Y. Zhan, and B. Du, "Token contrast for weakly-supervised semantic segmentation," 2023, *arXiv:2303.01267*.

[26] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020, pp. 1–22.

[27] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[28] L. Ru, H. Xu, Y. Xu, K. Huang, and Q. Tian, "Weakly supervised segmentation with stronger augmentations and coarse-to-fine pseudo labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 6974–6983.

[29] S.-H. Yoon, H. Kwon, H. Kim, and K.-J. Yoon, "Class tokens infusion for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3595–3605, doi: 10.1109/CVPR52733.2024.00345.

[30] F. Yang et al., "ConvFormer-CD: Hybrid CNN–transformer with temporal attention for detecting changes in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614815, doi: 10.1109/TGRS.2025.3544651.

[31] Y. Chen et al., "SparseFormer: A credible dual-CNN expert-guided transformer for remote sensing image segmentation with sparse point annotation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[32] T. Guo, R. Wang, F. Luo, X. Gong, L. Zhang, and X. Gao, "Dual-view spectral and global spatial feature fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, May 2023, Art. no. 5512913.

[33] Y. Duan, A. Li, F. Luo, Z. Jing, Y. Niu, and Y. Li, "Clustering hypergraph U-Net with high-order topology-aware pooling and unpooling for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[34] P. Krähenbühl and V. Koltun, "Efficient inference in fully connected CRFs with Gaussian edge potentials," in *Proc. 24th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, Red Hook, NY, USA, 2011, pp. 109–117.

[35] J.-C. Lee, D. Kim, and H. Shim, "Anti-adversarially manipulable weakly-supervised semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2021, pp. 1674–1683.

[36] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5574–5584.

[37] Z. Yang, K. Fu, M. Duan, L. Qu, S. Wang, and Z. Song, "Separate and conquer: Decoupling co-occurrence via decomposition and representation for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3606–3615.

[38] Y. Su, R. Sun, G. Lin, and Q. Wu, "Context decoupling augmentation for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6984–6994.

[39] F. Zhang, C. Gu, C. Zhang, and Y. Dai, "Complementary patch for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7242–7251.

[40] Y. Wang, Y. Zhou, X. Chen, S. Bai, and C. Huang, "Semi-supervised semantic segmentation using reliable pseudo-labels and consistency regularization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8124–8139, Nov. 2022.

[41] Y. Lin et al., "CLIP is also an efficient segmenter: A text-driven approach for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, 2023, pp. 15305–15314, doi: 10.1109/CVPR52729.2023.01469.

[42] Y. Sun, W. Li, F. Gao, and H. Zhang, "Decoupling semantic and spatial representations for weakly supervised semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[43] L. Xu, W. Ouyang, M. Bennamoun, F. Boussaid, and D. Xu, "Multi-class token transformer for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4310–4319.

[44] Y. Wei, J. Feng, X. Liang, M.-M. Cheng, Y. Zhao, and S. Yan, "Object region mining with adversarial erasing: A simple classification to semantic segmentation approach," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1568–1576.

[45] J. Feng, X. Wang, T. Li, S. Ji, and W. Liu, "Weakly-supervised semantic segmentation via online pseudo-mask correcting," *Pattern Recognit. Lett.*, vol. 165, pp. 33–38, Jan. 2023.

[46] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.

[47] *Isprs.2D Semantic Labeling Contest-Vaihingen*. Accessed: Dec. 1, 2024. [Online]. Available: http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html

[48] M. Everingham. (2009). *The Pascal Visual Object Classes Challenge 2007*. [Online]. Available: http://www.pascal-network.org/challenges/VOC/voc2007/workshop/index.html

[49] B. Hariharan, P. Arbeláez, L. Bourdev, S. Maji, and J. Malik, "Semantic contours from inverse detectors," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 991–998.

[50] Y. Wang, J. Zhang, M. Kan, S. Shan, and X. Chen, "Self-supervised equivariant attention mechanism for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12272–12281.

[51] J. Fan, Z. Zhang, T. Tan, C. Song, and J. Xiao, "CIAN: Cross-image affinity net for weakly supervised semantic segmentation," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, 2020, pp. 10762–10769.

[52] Y.-T. Chang, Q. Wang, W.-C. Hung, R. Piramuthu, Y.-H. Tsai, and M.-H. Yang, "Weakly-supervised semantic segmentation via sub-category exploration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8988–8997.

[53] W. Liu, X. Kong, T.-Y. Hung, and G. Lin, "Cross-image region mining with region prototypical network for weakly supervised segmentation," *IEEE Trans. Multimedia*, vol. 25, pp. 1148–1160, 2023.

[54] R. Zhou et al., "Weakly supervised semantic segmentation in aerial imagery via cross-image semantic mining," *Remote Sens.*, vol. 15, no. 4, p. 986, Feb. 2023, doi: 10.3390/rs15040986.

[55] R. Zhou et al., "Weakly supervised semantic segmentation in aerial imagery via explicit pixel-level constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[56] G. Papandreou, L.-C. Chen, K. P. Murphy, and A. L. Yuille, "Weakly-and semi-supervised learning of a deep convolutional network for semantic image segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1742–1750.

[57] P. O. Pinheiro and R. Collobert, "From image-level to pixel-level labeling with convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1713–1721.

[58] A. Roy and S. Todorovic, "Combining bottom-up, top-down, and smoothness cues for weakly supervised image segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7282–7291.

[59] B. Zhang, J. Xiao, Y. Wei, M. Sun, and K. Huang, "Reliability does matter: An end-to-end weakly supervised semantic segmentation approach," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 12765–12772.

[60] X. Zhang et al., "Adaptive affinity loss and erroneous pseudo-label refinement for weakly supervised semantic segmentation," in *Proc. 29th ACM Int. Conf. Multimedia*, Oct. 2021, pp. 5463–5472.

[61] Z. Yang, X. Zhao, X. Wang, Q. Zhang, and J. Xiao, "FFR: Frequency feature rectification for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 30261–30270.

[62] P. Jiang, Q. Hou, Y. Cao, M. Cheng, Y. Wei, and H. Xiong, "Integral object mining via online attention accumulation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 2070–2079.

[63] G. Sun, W. Wang, J. Dai, and L. Van Gool, "Mining cross-image semantics for weakly supervised semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, Glasgow, U.K. Cham, Switzerland: Springer, 2020, pp. 347–365.

[64] L. Xu, W. Ouyang, M. Bennamoun, F. Boussaid, F. Sohel, and D. Xu, "Leveraging auxiliary tasks with affinity learning for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6984–6993.

[65] Y. Yao et al., "Non-salient region object mining for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2623–2632.

[66] T. Wu et al., "Embedded discriminative attention mechanism for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16765–16774.

[67] S. Lee, M. Lee, J. Lee, and H. Shim, "Railroad is not a train: Saliency as pseudo-pixel supervision for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5495–5505.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.

[68] B. Zhang, J. Xiao, J. Jiao, Y. Wei, and Y. Zhao, "Affinity attention graph neural network for weakly supervised semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8082–8096, Nov. 2022.



Ruixue Zhou received the bachelor's degree from Shandong University, Jinan, China, in 2018. She is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

She is interested in computer vision, pattern recognition, and remote sensing image processing, particularly weakly supervised semantic segmentation, scene understanding, and remote sensing image interpretation.



Jihao Li (Member, IEEE) received the B.Sc. degree in electronic information engineering from Xidian University, Xi'an, China, in 2017, and the Ph.D. degree in signal and information processing from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2022.

He is currently an Assistant Researcher with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image interpretation.



Wenkai Zhang (Member, IEEE) received the B.Sc. degree in electronic information engineering from China University of Petroleum, Qingdao, China, in 2013, and the M.Sc. and Ph.D. degrees in electronic information engineering from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2018.

He is currently an Assistant Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include remote sensing image semantic segmentation and multimodal information processing.



Shuoke Li received the B.Sc. degree from Xi'an Jiaotong University, Xi'an, China, in 2017, and the M.Sc. degree from Peking University, Beijing, China, in 2020.

He is currently an Assistant Engineer with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include multimodal signal processing, image segmentation, and pattern recognition.



Jialiang Chen (Member, IEEE) received the B.Sc. degree from Zhengzhou University, Zhengzhou, China, in 2012, and the M.Sc. degree from Beijing Institute of Technology, Beijing, China, in 2016.

He is currently an Assistant Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include remote sensing image processing.



Chongyang Li received the B.Sc. degree in electronic information engineering from Dalian University of Technology, Dalian, China, in 2022. He is currently pursuing the Ph.D. degree in signal and information processing with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include multimodal remote sensing image interpretation and multimodal signal processing.



Boyuan Tong (Graduate Student Member, IEEE) received the B.Sc. degree in electronic information engineering from Central South University, Changsha, China, in 2022. He is currently pursuing the joint Ph.D. degree in communication and information system with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include computer vision and pattern recognition, especially in remote sensing image interpretation and multimodal signal processing.



Weihang Zhang (Graduate Student Member, IEEE) received the B.Sc. degree in electronic information engineering from Xidian University, Xi'an, China, in 2021. He is currently pursuing the Ph.D. degree in signal and information processing with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include remote sensing image retrieval and multimodal remote sensing image interpretation.



Xian Sun received the B.Sc. degree from Beijing University of Aeronautics, Beijing, China, in 2004, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2009.

Currently, he is a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests are computer vision, geospatial data mining, and remote sensing image understanding.



Kun Fu (Senior Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in electronic information engineering from the National University of Defense Technology, Changsha, China, in 1995, 1999, and 2002, respectively.

He currently works as a Professor at the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. He is interested in remote sensing image analysis, geospatial data mining, and visualization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:45 UTC from IEEE Xplore. Restrictions apply.